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# **Comparing Different GARCH Models with Respect to Financial Risk Forecasting**

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# 1 Introduction

Value at Risk (VaR) and Expected Shortfall (ES) are widely accepted and used measurements to assess risk of financial investments and are key metrics for risk management. The Basel Committee on Banking Supervision, an international regulatory body that sets global standards for the regulation, supervision, and risk management of banks, sets them as the standard measurements for the assessment of market risk. They are an important part of the regulatory framework that governs the calculation of capital requirements for banks (Basel Committee on Banking Supervision, 2017).

Different methods for the calculation of these risk measurements have been discussed in the literature. GARCH-type models, initially introduced by Engle (1982) and Bollerslev (1986), with different GARCH specifications and innovation distribution assumptions, have particularly proven themselves to be an appropriate way to forecast the conditional standard deviation, a key component of VaR and ES estimates. They are able to capture often observed features of return series like volatility clustering, persistence of volatility, dynamic response to past shocks and, depending on their formulation, also asymmetric volatility. Because VaR has been introduced earlier than ES, there is a substantial body of research on the best GARCH-type models for VaR estimation. In contrast, the number of studies examining the optimal GARCH-type model for ES remains comparatively smaller.

Since the results regarding the best GARCH specification and distribution assumption for VaR and ES estimation vary in the literature, I tested various models on different asset classes for this study. These asset classes include stock indices, represented by the DAX and WIG, a cryptocurrency, represented by Bitcoin prices, and a commodity, represented by Gold spot prices. The goal has been to identify key characteristics that GARCH specifications and distribution assumptions must have to yield the best-performing models for one-step-ahead VaR and ES forecasting for each asset class, while also exploring potential differences between asset classes. Additionally, I aimed to gain insights into the relative importance of the GARCH specification versus the distribution assumption of a model. After estimating one-step-ahead VaR and ES forecasts for each considered model using a rolling-window approach, I assessed the validity of these forecasts through two conditional coverage backtests introduced by Christoffersen (1998) and Du and Escanciano (2015). Subsequently, I ranked the models that passed these backtests using the FZ loss function, which was introduced by Fissler and Ziegel (2016).

The paper is structured as follows: Firstly, I will start with a literature review and describe the data used for this study. Then, I will explain the methodology. Finally, I will present and explain the results of different models and come to a conclusion.

## 2 Literature Review

The objective of many papers is either to propose new models based on theoretical considerations or to compare different model specifications and distribution assumptions in their empirical performance. In older papers, various model specifications are often the primary focus. Studies that consider more sophisticated distribution assumptions beyond the Normal and Student's  $t$ -distribution have only started to emerge more recently.

Gokcan (2000), for example, compared the performance of a linear GARCH(1,1) model against a non-linear EGARCH(1,1) model for volatility forecasts from stock market returns of multiple countries. For the comparison of the in-sample fit of these models, he used the Akaike Information Criterion. For the out-of-sample fit, he used the Mean Squared Error. He found that the GARCH(1,1) model outperformed the EGARCH(1,1) model in out-of-sample forecasting, even though the return series exhibited asymmetric volatility.

The effectiveness of various GARCH-type models with different conditional mean equations, changing symmetrical distribution assumptions, and different sample sizes in forecasting VaR of diversified portfolios across different stock markets, was investigated by Angelidis et al. (2004). They found that leptokurtic distribution assumptions produce better VaR forecasts than the Normal distribution, especially in cases of extreme market movements. Smaller sample sizes, between 250 and 500 consecutive return observations, have proven to lead to more accurate one-step-ahead VaR forecasts than larger sample sizes. The exact choice of the conditional mean equation does not change the forecasting accuracy significantly, and more complex equations usually only led to more parameters that have to be estimated, which is not desired in the spirit of model parsimony. The overall best-performing model in their study was the EGARCH model with the Student's  $t$ -distribution, but results may vary from asset to asset.

Awartani and Corradi (2005) examined how well various sophisticated GARCH model specifications predict out-of-sample volatility of the S&P500, particularly focusing on how the ability to model asymmetric volatility improves forecast precision compared to the forecasts from the standard GARCH(1,1) model. Their findings were that asymmetric GARCH models outperform the standard GARCH(1,1) model in one-step-ahead volatility prediction, because they are able to model the leverage effect. Compared to other symmetric GARCH models, the GARCH(1,1) model performed the best.

Hansen and Lunde (2005) underline that the best performing model depends on the market where it is applied to and that one should carefully weigh out between flexibility and parsimony. In their study, they aimed to check whether more advanced volatility models

provide better one-day-ahead volatility predictions than the standard GARCH(1,1) model. Their conclusion was that for the prediction of stock market volatility, asymmetric models led to significantly better predictions than the standard GARCH(1,1) model. For the DM / USD exchange rate, however, the standard GARCH(1,1) model performed better.

Orhan and Köksal (2012) tested 16 different GARCH model specifications, each with Normal and Student's t-distribution and a MA(1) term for the conditional mean equation, against each other in forecasting the VaR of four stock indices, two from emerging economies and two from developed economies. As a backtesting scheme, they used the unconditional coverage test from Kupiec (1995) and the conditional coverage test from Christoffersen (1998). For the comparison of the models, they used a quadratic loss function. The ARCH models were often superior to more complicated model specifications. The Student's t-distribution was only slightly better than the Normal distribution.

Whether there is a pattern regarding the quality of different models and methods for ES estimation of distinct asset classes was investigated by Righi and Ceretta (2015). Different unconditional, conditional and quantile or expectile regression-based models with various skewed and symmetric distribution assumptions were examined. They tested three estimation window lengths: 250, 500 and 1000. For backtesting ES estimates, they used a test developed by McNeil and Frey (2000) and proposed an ES backtest themselves. Their findings were that conditional volatility models, namely GARCH-type models, and filtered historical simulation methods generally perform the best. Contrary to Angelidis et al. (2004), they concluded that longer estimation windows should usually be preferred, since shorter windows tend to produce inferior ES estimation results. They pointed out the importance of respecting stylized facts of financial returns like volatility clustering, heavy tails, asymmetry and skewness in order to arrive at realistic risk estimates.

Sobreira and Louro (2020) found no significant improvement when including skewed distributions for forecasting the VaR and ES for the Portuguese stock market. In their work, they compared a broad set of GARCH specifications in combination with different distribution assumptions, namely the Normal distribution, Student's t-distribution and Generalized Error Distribution, and their skewed versions. Asymmetric GARCH models, in combination with Extreme Value Theory (EVT), generally performed the best. Changing the symmetric distributions into their skewed version did not significantly improve the model forecasts and only added additional parameters that had to be estimated.

A study that specifically investigates the relative importance of distribution assumptions compared to GARCH specifications in volatility models for financial returns was conducted by Garcia-Jorcano and Novales (2021). By comparing different GARCH model specifications with a broad set of symmetric and asymmetric distribution assumptions,



their aim was to determine which of these two factors plays a more important role in volatility forecasting. Their findings were that the distribution assumption plays a more significant role for accurate volatility forecasting compared to the exact GARCH specification, although GARCH specifications that incorporate asymmetry tend to perform better than symmetric specifications. Skewed distributions performed generally better than their symmetric counterparts.

The literature review reveals that while various GARCH specifications and distribution assumptions have been extensively studied in terms of their effectiveness regarding volatility and financial risk forecasting, no single specification or distribution universally outperforms all others. In many studies, asymmetric specifications indeed performed better than symmetrical ones, but there are also enough examples where symmetric specifications were better. In most cases, assuming leptokurtic innovation distributions led to better risk estimates. Additionally including the possibility of modeling skewness, only partially improved the models and sometimes only led to more parameters that had to be estimated. These insights underline the importance of evaluating multiple models to identify the most suitable one for a given context.

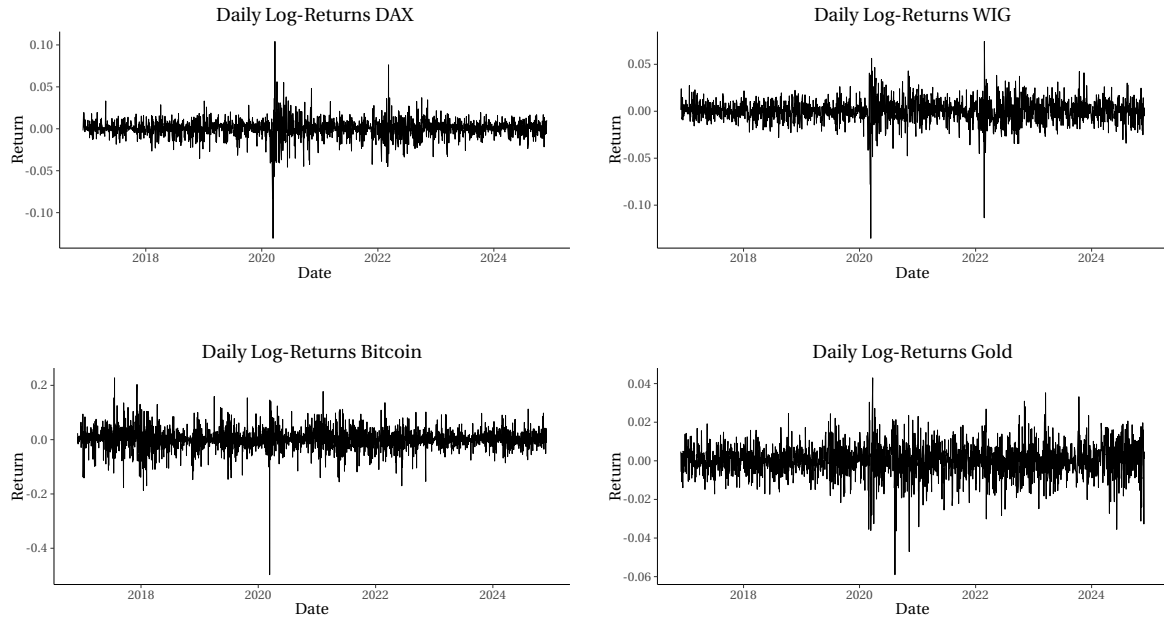
## **3 Data Description**

### **3.1 Data Sets**

To gain a comprehensive view on the best-performing GARCH-type model for VaR and ES forecasting of different asset classes and to identify potential differences between these asset classes, I used four indices that represent three different asset classes. These indices include two stock market indices. The first one is the DAX, which represents the 40 largest blue chip companies listed on the Frankfurt Stock Exchange in Germany (Finance, 2024a). The second stock market index is the WIG. It is the main index of the Warsaw Stock Exchange and comprises all companies listed on the stock exchange in Poland (Finance, 2024b). I sourced data for both indices from Google Finance by using the `GOOGLEFINANCE` function in Google Sheets. In addition to these two stock indices, I included a series of historical Bitcoin prices (Investing.com, 2024a) as well as historical Gold spot prices (Investing.com, 2024b) in this study, to cover a popular cryptocurrency and a commodity. Both asset prices are denoted in US dollars. I sourced data for them from Investing.com.

All data sets consist of daily prices or index points, which I transformed into daily log-returns. They cover an eight-year period starting from 1st December 2016 to 30th November 2024. This period encompasses times of financial turmoil, such as the outbreak of the

COVID-19 pandemic in 2020 and the start of the war in Ukraine in 2022, as well as times with more stable economical conditions. This diverse time frame challenges the models in different market conditions and tests their robustness across varying economic circumstances. With 2032 observation for DAX, 2000 for WIG, 2922 for Bitcoin and 2082 for Gold, the eight-year period provides a sufficient number of data points to accurately estimate and test various risk models and compare their the performance.



**Figure 3.1: Daily Log-Return Plots**

Figure 3.1 shows the daily log-return series for all four indices. Corresponding price series plots can be found in Figure A.1. It can be seen that log-returns moved around 0. Furthermore, volatility was not constant over time. Rather, each index had times with higher volatility and times with lower volatility. Especially at the beginning of 2020, at the time the COVID-19 pandemic broke out, it can be seen that volatility surged notably across all four indices. Both stock market index plots additionally indicate a significant peak in volatility at the beginning of 2022, when the war in Ukraine started. These changes of volatility indicate that the standard deviation of log-returns is not constant over time, which is a commonly observed feature of financial time series (Cont, 2007). It substantiates the usage of models which represent the standard deviation as a function of preceding volatility. GARCH-type models, examined in this study, are a prominent example of such kind of models.

The following subsection provides further insides into the data with a more detailed statistical description and serves as a foundation to justify the modeling approach I took for this study.

### 3.2 Statistical Description

|     | Mean   | Median | S.D.   | Skewness | Kurtosis | JB Test p | LB Test p | Arch Test p |
|-----|--------|--------|--------|----------|----------|-----------|-----------|-------------|
| DAX | 0.0003 | 0.0007 | 0.0116 | -0.6967  | 17.9836  | 0.0000    | 0.0000    | 0.0000      |
| WIG | 0.0002 | 0.0004 | 0.0123 | -1.0863  | 16.3340  | 0.0000    | 0.1302    | 0.0000      |
| BTC | 0.0017 | 0.0014 | 0.0381 | -0.7949  | 16.0279  | 0.0000    | 0.0275    | 0.0000      |
| GLD | 0.0004 | 0.0006 | 0.0085 | -0.3712  | 6.3121   | 0.0000    | 0.2737    | 0.0000      |

**Table 3.1: Statistical Description**

In Table 3.1 I report central descriptive statistics and  $p$ -values of tests that are commonly used to gain a comprehensive picture of the characteristics of a time series. Furthermore, additional descriptive plots of all four indices with short explanations can be found in the appendix (Figures A.2, A.3, A.4 & A.5).

The log-return series of all indices have a mean and median which are positive, but close to 0. With 0.0017 and 0.0014, Bitcoin has the highest mean and median. The mean and median of the other three log-return series are all in a range between 0.0002 and 0.0007. For all indices, apart from Bitcoin, the mean log-returns are smaller than the median log-returns. In combination with a small negative skewness coefficient between  $-0.3712$  for Gold and  $-1.0863$  for WIG, it can be deduced that the log-return densities of all four indices are weakly left skewed. This characteristic can additionally be seen in the density plots in Figures A.2, A.3, A.4 and A.5. The empirical standard deviation is with 0.0381 the highest for Bitcoin log-returns. The WIG and DAX log-returns are with 0.0123 and 0.0116 in the middle field and Gold log-returns have with 0.0085 the lowest empirical standard deviation. As the standard deviation serves as a direct input for the calculation of VaR and ES, a higher standard deviation leads, on average, to higher VaR and ES estimates in absolute terms and therefore higher assumed risk, as long as the higher standard deviation is not offset by a higher mean. All four log-return series exhibit a kurtosis which is greater than 3, the kurtosis of the Normal distribution. Higher kurtosis in log-return series indicates that the distribution has fatter tails and a sharper peak compared to the Normal distribution, implying a greater likelihood of extreme returns and more concentrated moderate returns. Gold log-returns have with 6.3121 the lowest kurtosis. DAX, WIG and Bitcoin log-returns have with 17.9836, 16.3340 and 16.0279 a notably higher kurtosis. This can additionally be seen in the density plots in Figures A.2, A.3, A.4 and A.5. Looking at Figures A.3a and A.5a, the distributions of WIG and Gold log-returns can be relatively well approximated by a Student  $t$ -distribution with 5 degrees of freedom. Figures A.2a, A.3a, A.4a and A.5a show that the Normal distribution does not lead to a good approxima-

tion of the empirical distribution of all four log-return series.

For testing the normality of the observed log-return series statistically, I employed the Jarque–Bera test which has normality as its  $H_0$ . In Table 3.1, I report the  $p$ -values of the test results. The  $p$ -values for all log-return series are 0 and therefore, I can reject the  $H_0$  and can deduce that the Normal distribution is not a good approximation for the raw log-returns at hand. This can give a hint that the Normal distribution is not the distribution the innovations of the later build models exhibit, but it does not finally exclude the Normal distribution, as the distribution of innovations can differ from the distribution of raw log-returns.

To further examine the dynamics of the given log-return series, more specifically their serial dependencies, I employed the Ljung–Box test which has autocorrelation coefficients of 0 for a specified number of lags as its  $H_0$ . I used 10 lags for this test. The results of the tests show that I cannot reject the  $H_0$  for WIG and Gold log-return series which means that the expected values of log-returns for this indices seem to be serially uncorrelated. This suggests, that a simple intercept model is sufficient for the mean model. However, for DAX and Bitcoin, I have to reject the  $H_0$  of serial uncorrelated expected values ( $p < 0.05$ ). To examine these significant autocorrelation coefficients further, one can have a look at autocorrelation (ACF) and partial autocorrelation (PACF) plots. For DAX and Bitcoin, these plots are presented in Figures A.2e, A.2f, A.4e and A.4f. The ACF and PACF plots of the DAX reveal that the autocorrelation and partial autocorrelation coefficients are small, with absolute values lower than 0.08, and they go into different directions, as some are negative while others are positive. This pattern appears random and does not provide any meaningful information. Consequently, I assume that a simple intercept mean model is sufficient as well. The same goes for Bitcoin log-returns (Figures A.4e & A.4f). Although I want to point out that here, the first two lags exhibit significant autocorrelation and partial autocorrelation coefficients, which might be valuable information. However, these coefficients are very small, and the additional information they provide is minimal. While applying a simple intercept model for the expected value is not as straightforward here as for the other indices, it can still be justified. This is particularly true given that the focus of this study is on volatility models, and financial theory, in line with the concept of market efficiency, suggests that the expected values of returns are independent of preceding returns (Fama, 1970). The found evidence against this theory is not clear enough to disprove this assumption. Hence, a simple intercept model as the mean model is appropriate for all four log-return series.

In order to statistically test for serial dependencies in volatility, and therefore the suspicion of volatility clustering already raised in the previous subsection based on the log-

return series plots in Figure 3.1, I employed the ARCH test. The  $H_0$  of this test is that squared log-returns have autocorrelation coefficients of zero for a specified number of lags. For this study, I used 10 lags.  $p$ -values for all four series are 0. Accordingly, I can reject the  $H_0$ . All four indices show volatility clustering in their log-return series. This picture gets even more clear when looking on the ACF and PACF plot of squared returns presented in Figures A.2, A.3, A.4 and A.5. Based on these findings, I can say that dynamically modeling the standard deviation depending on previous volatility is necessary for reliable VaR and ES estimation.

Lastly, to check for asymmetric volatility, I used cross correlation plots (CCF) between squared log-returns and lagged log-returns, which I report in Figure A.2d, A.3d, A.4d and A.5d. Both stock market indices have significant negative cross correlation coefficients over multiple lags, meaning that negative log-returns are often followed by higher volatility compared to positive log-returns. In a weaker form, this can also be observed for Bitcoin. In the CCF plot for Gold log-returns, this feature does not appear.

In the following sections, I will build upon these findings to describe the construction of the tested models and the tests used to validate and compare them.

## 4 Methodology

### 4.1 Log-Return Process

Before introducing the risk measures examined in this study and the models used to estimate them, I want to start by setting up a theoretical framework regarding the assumed log-return process and mention its theoretical and often observed properties in comparison to properties the examined indices of this study exhibit. In the financial literature, log-returns are often preferred over normal returns, because they are additive over time (Fama, 1970). This characteristic simplifies financial modeling and statistical analysis. Daily log-returns, examined in this study, are defined as follows:

$$\begin{aligned} r_t &= \log\left(\frac{P_t}{P_{t-1}}\right) = \log(P_t) - \log(P_{t-1}) \\ r_t &= \mu + \sigma_t \epsilon_t \\ \epsilon_t &\sim d(\theta) \end{aligned} \tag{4.1}$$

where  $P_t$  is the closing price at time  $t$ ,  $r_t$  is the daily log-return at time  $t$ ,  $\log()$  represents the natural logarithm,  $\epsilon_t$  is an innovation term at time  $t$ ,  $d$  is a general standardized den-

sity function (pdf) with a general parameter vector  $\theta$  where the expected value is 0 and the variance is 1.

These log-returns commonly exhibit certain characteristics, in the financial literature often regarded as stylized facts. Theoretical models modeling financial risk, with VaR and ES as popular examples, are based on these characteristics. This is why it is imported to mention them and check if they are apparent in the log-return series examined in this study. The first characteristic is the absence of serial autocorrelation, stemming from a weak form of market efficiency (Fama, 1970). As identified in section 3, this can be observed in the case of WIG and Gold log-returns, and following the presented argumentation, I also assume this for DAX and Bitcoin log-returns for this study. Therefore, I assume a constant  $\mu$ . Furthermore, returns often show volatility clustering, which is very apparent in the data at hand. It introduces the need to model the standard deviation  $\sigma_t$  dynamically based on preceding volatility. Volatility often reacts asymmetrically to shocks. It often gets stronger after negative shocks compared to positive shocks. This characteristic is called leverage effect and introduces the need to apply models which account for this feature. In the used data, this trait can be observed for the DAX and the WIG, represented by negative cross correlation coefficients between squared log-returns and lagged log-returns, and in a weaker form for Bitcoin log-returns, but not for Gold log-returns. Another commonly observed feature is the non-normality of returns, more specifically, leptokurtic densities (excess kurtosis) due to a higher probability of extreme events compared to the Normal distribution, and negative skewness, due to asymmetric gain and loss intensities. Both characteristics can be observed for all four indices and have the implication that innovations  $\epsilon_t$  could also exhibit this feature, and therefore more sophisticated distributions  $d$  than the Normal distribution might be needed for modeling financial risk for these indices.

## 4.2 Theoretical Foundation of Value at Risk and Expected Shortfall

As the first measure of financial risk for this study, I used the Value at Risk. The VaR is defined as the minimum return that is expected with a predefined probability, known as the tolerance level  $\alpha$ . In other words, the probability of observing a return lower than  $Var_t^\alpha$  at a given time  $t$  is  $\alpha$ . In mathematical terms, the  $Var_t^\alpha$  is defined as follows:

$$\begin{aligned} P[r_t \leq Var_t^\alpha] &= \alpha \\ Var_t^\alpha &= \mu + \sigma_t q_\alpha \\ q_\alpha &= D^{-1}(\alpha) \end{aligned} \tag{4.2}$$

where  $D$  is the distribution function (cdf) of  $d$ . Consequently, the inverse cdf,  $D^{-1}(\alpha)$ , calculates the  $\alpha$ -quantile  $q_\alpha$  of  $d$ .

Since  $Var_t^\alpha$  is a located and scaled quantile of  $d$ , it can also be expressed as a quantile of a located and scaled version of  $d$ , called  $f_t$ :

$$Var_t^\alpha = F_t^{-1}(\alpha) \quad (4.3)$$

where  $f_t$  is time-dependent due to its time-dependent scaling parameter  $\sigma_t$ .

The VaR was the primary risk measure for market risk before Basel III (Basel Committee on Banking Supervision, 2006). Due to its theoretical deficiencies, it has been replaced by the second financial risk measure that I used for this study, the Expected Shortfall, which is the primary market risk measure as of Basel III (Basel Committee on Banking Supervision, 2019). It is defined as the expected value of return  $r_t$  at a given time  $t$ , if return  $r_t$  is lower than the  $Var_t^\alpha$ :

$$ES_t^\alpha = E[r_t \mid r_t \leq Var_t^\alpha] \quad (4.4)$$

Following Du and Escanciano (2015), and using equation 4.3, the  $ES_t^\alpha$  can be represented in terms of  $Var_t^\alpha$ :

$$\begin{aligned} ES_t^\alpha &= \frac{1}{\alpha} \int_0^\alpha F_t^{-1}(u) du \\ &= \frac{1}{\alpha} \int_0^\alpha Var_t^u du \end{aligned} \quad (4.5)$$

Using the definition of  $Var_t^\alpha$  from equation 4.2 yields the following representation of  $ES_t^\alpha$ :

$$\begin{aligned} ES_t^\alpha &= \mu + \sigma_t \frac{1}{\alpha} \int_0^\alpha q_u du \\ q_u &= D^{-1}(u) \end{aligned} \quad (4.6)$$

One advantage of the ES over the VaR as a financial risk measure is that the ES better captures tail risk. This is because in contrast to VaR, the ES does not only take one point of the distribution into account. It utilizes the whole lower tail. This means that VaR only provides information about the maximum loss that has to be expected with a certain probability  $\alpha$ . The ES on the other hand also provides information about the expected magnitude of loss that occurs with a probability of  $\alpha$ . Therefore, it also measures the severity of loss (Du

and Escanciano, 2015). Using the whole lower tail of the distribution also means that the distribution assumption might play an even bigger role in ES estimation compared to VaR estimation.

Furthermore, in contrast to VaR, ES possesses all the characteristics of a coherent risk measure, namely monotonicity, subadditivity, positive homogeneity and translation invariance, which are described by Artzner et al. (1999). VaR, in particular, lacks sub-additivity (Artzner et al., 1999), which can lead to an overestimation of risk when aggregating individual VaR estimates across multiple assets and thereby mislead the risk assessment.

Despite the deficiencies of VaR compared to ES, I included both risk measures in this study to gain a broader perspective on appropriate models for financial risk measurement of the examined asset classes. This approach allows for a comprehensive evaluation of the best-performing models for estimating the components of both measures.

### 4.3 Empirical Prediction of Value at Risk and Expected Shortfall

To estimate predictions for  $Var_{T+1}^\alpha$  and  $ES_{T+1}^\alpha$  at time  $T$  in a parametric way, it is necessary to have estimates for three different components. These are the predicted mean log-return  $\hat{\mu}_{T+1}$ , the predicted standard deviation  $\hat{\sigma}_{T+1}$  and the assumed innovation distribution  $d$  with its estimated parameter vector  $\hat{\theta}$ .

Referring back to section 3 and 4.1, I have already worked out that due to very weak to no autocorrelation in all log-return series, in combination with the efficient market hypothesis, an intercept mean model is appropriate for the examined data. This means that  $\hat{\mu}_{T+1}$  is equal to a constant estimated mean  $\hat{\mu}$  without any autoregressive (AR) or moving average (MA) term.

$$\hat{\mu}_{T+1}(\tilde{\Omega}_T) = \hat{\mu} \quad (4.7)$$

Due to volatility clustering in the log-returns of all indices, the predicted standard deviation  $\hat{\sigma}_{T+1}$  is calculated as a function  $V$  of preceding volatility measures, returns, and innovations summarized in  $\tilde{\Omega}_T$  and parameterized by an estimated parameter vector  $\hat{\Psi}$ .  $V$  represents different GARCH-type models:

$$\hat{\sigma}_{T+1}(\tilde{\Omega}_T) = V(\hat{\Psi}, \tilde{\Omega}_T) \quad (4.8)$$

The last component is the assumed distribution of innovations  $d$  with its estimated parameter vector  $\hat{\theta}$  which, by definition, has a location parameter of 0 and a scaling param-



eter of 1. Additionally, it can also include parameters for skewness and shape, depending on the definition of the assumed distribution.

$\hat{\mu}$ ,  $\hat{\Psi}$  and  $\hat{\theta}$  are estimated with changing functional representations  $V$  of the standard deviation and changing innovation distribution assumptions  $d$  using the maximum likelihood estimation approach:

$$\begin{aligned} L(\mu, \Psi, \theta \mid \tilde{\Omega}_T) &= \prod_{t=1}^T d(\mu, V(\Psi, \cdot), \theta \mid \tilde{\Omega}_T) \\ \ell(\mu, \Psi, \theta \mid \tilde{\Omega}_T) &= \log(L(\mu, \Psi, \theta \mid \tilde{\Omega}_T)) \\ (\hat{\sigma}, \hat{\Psi}, \hat{\theta}) &= \arg \max_{\mu, \Psi, \theta} \ell(\mu, \Psi, \theta \mid \tilde{\Omega}_T) \end{aligned} \quad (4.9)$$

To finally receive  $Var_{T+1}^\alpha$  and  $ES_{T+1}^\alpha$  in time  $T$ , the following calculations apply:

$$\begin{aligned} Var_{T+1}^\alpha(\tilde{\Omega}_T) &= \hat{\mu} + \hat{\sigma}_{T+1}(\tilde{\Omega}_T) D^{-1}(\hat{\theta}, \alpha) \\ ES_{T+1}^\alpha(\tilde{\Omega}_T) &= \hat{\mu} + \hat{\sigma}_{T+1}(\tilde{\Omega}_T) \frac{1}{\alpha} \int_0^\alpha D^{-1}(\hat{\theta}, u) du \end{aligned} \quad (4.10)$$

#### 4.4 GARCH-Type Specifications

The goal of the empirical study is to work out the best-performing function  $V$  to predict  $\hat{\sigma}_{t+1}$ , in combination with the best-performing innovation distribution assumption  $d$ , to calculate both introduced financial risk measures for different asset classes. For  $V$ , the focus lies on GARCH-type models. GARCH-type models (Generalized Autoregressive Conditional Heteroskedasticity models) are widely used in finance to model and predict time-varying volatility of financial time series. Depending on their formulation, they are capable of modeling key characteristics of financial time series such as volatility clustering, persistence of volatility, dynamic response to past shocks and asymmetric volatility. The following table presents all GARCH-type models examined in this study:

| Specification       | Introduced by                  | Functional Representation $V$ (Ghalanos, 2023)                                                                                                                                                       |
|---------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARCH( $q$ )         | Engle (1982)                   | $\sigma_t^2 = \alpha_0 + \sum_{j=1}^q \alpha_j \epsilon_{t-j}^2$                                                                                                                                     |
| GARCH( $q, p$ )     | Bollerslev (1986)              | $\sigma_t^2 = \omega + \sum_{j=1}^q \alpha_j \epsilon_{t-j}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$                                                                                                 |
| EGARCH( $q, p$ )    | Nelson (1991)                  | $\log(\sigma_t^2) = \omega + \sum_{j=1}^q (\alpha_j z_{t-j} + \gamma_j ( z_{t-j}  - \sqrt{\frac{2}{\pi}})) + \sum_{j=1}^p \beta_j \log(\sigma_{t-j}^2)$                                              |
| GJR-GARCH( $q, p$ ) | Glosten et al. (1993)          | $\sigma_t^2 = \omega + \sum_{j=1}^q (\alpha_j + \gamma_j \mathbb{I}(\epsilon_{t-j} \leq 0)) \epsilon_{t-j}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$                                                  |
| TGARCH( $q, p$ )    | Zakoian (1994)                 | $\sigma_t = \omega + \sum_{j=1}^q \alpha_j ( \epsilon_{t-j}  - \gamma_j \epsilon_{t-j}) + \sum_{j=1}^p \beta_j \sigma_{t-j}$                                                                         |
| AVGARCH( $q, p$ )   | Taylor (1986) & Schwert (1990) | $\sigma_t = \omega + \sum_{j=1}^q \alpha_j  \epsilon_{t-j}  + \sum_{j=1}^p \beta_j \sigma_{t-j}$                                                                                                     |
| APARCH( $q, p$ )    | Ding et al. (1993)             | $\sigma_t^\delta = \omega + \sum_{j=1}^q \alpha_j ( \epsilon_{t-j}  - \gamma_j \epsilon_{t-j})^\delta + \sum_{j=1}^p \beta_j \sigma_{t-j}^\delta$                                                    |
| NGARCH( $q, p$ )    | Higgins and Bera (1992)        | $\sigma_t^\lambda = \omega + \sum_{j=1}^q \alpha_j  \epsilon_{t-j} ^\lambda + \sum_{j=1}^p \beta_j \sigma_{t-j}^\lambda$                                                                             |
| NAGARCH( $q, p$ )   | Engle and Ng (1993)            | $\sigma_t^2 = \omega + \sum_{j=1}^q \alpha_j \sigma_{t-j}^2 ( z_{t-j} - \eta_{2j} )^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$                                                                         |
| FGARCH( $q, p$ )    | Hentschel (1995)               | $\sigma_t^\lambda = \omega + \sum_{j=1}^q \alpha_j \sigma_{t-j}^\lambda ( z_{t-j} - \eta_{2j}  - \eta_{1j}(z_{t-j} - \eta_{2j}))^\delta + \sum_{j=1}^p \beta_j \sigma_{t-j}^\lambda$                 |
| CGARCH( $q, p$ )    | Lee and Engle (1993)           | $\sigma_t^2 = q_t + \sum_{j=1}^q \alpha_j (\epsilon_{t-j}^2 - q_{t-j}) + \sum_{j=1}^p \beta_j (\sigma_{t-j}^2 - q_{t-j})$<br>$q_t = \omega + \rho q_{t-1} + \phi(\epsilon_{t-1}^2 - \sigma_{t-1}^2)$ |

Table 4.1: Examined GARCH Specifications

The baseline GARCH-type models include the ARCH( $q$ ) model, which models the conditional variance based on  $q$  lagged squared errors, and the GARCH( $q, p$ ) model, which extends ARCH( $q$ ) by also incorporating  $p$  lagged conditional variance terms. Both models can capture symmetric volatility clustering, with the GARCH( $q, p$ ) model further decomposing volatility into short-term shock reactions and a more persistent component. All following GARCH-type models are in essence modifications and extensions of the standard GARCH( $q, p$ ) model.

In the EGARCH( $q, p$ ) model (Exponential GARCH), the conditional variance of the standard GARCH( $q, p$ ) model is replaced by the natural logarithm of the conditional variance, and the ARCH term, which captures short-term shocks, is extended by further decomposing it into a sign effect component ( $\alpha_j z_{t-j}$ ) and a size effect component ( $\gamma_j (|z_{t-j}| - \sqrt{\frac{2}{\pi}})$ ), where  $z_{t-j}$  is the standardized error of time  $t - j$ . This decomposition allows the model to capture asymmetric volatility, meaning it differentiates between the impact of positive and negative shocks on future volatility, if  $\alpha_j \neq 0$ . In the GJR-GARCH( $q, p$ ) model, an indicator function  $\mathbb{I}(\epsilon_{t-j} \leq 0)$  is added to the ARCH term of the standard GARCH( $q, p$ ) model that is 1 if  $\epsilon_{t-j} \leq 0$  and 0 otherwise. This means that the ARCH coefficient is  $\alpha_j$  if the  $j$ -times lagged error is positive, and  $\alpha_j + \gamma_j$  otherwise. Therefore, in cases where  $\gamma_j \neq 0$ , the conditional variance is modeled asymmetrically. In the equation of the TGARCH( $q, p$ ) model (Threshold GARCH), the conditional variance is replaced by the conditional standard deviation and absolute errors are used instead of squared errors.  $\gamma_j \epsilon_{t-j}$  introduces the possibility of modeling asymmetric volatility into the model, if  $\gamma_j \neq 0$ . In the AVGARCH( $q, p$ ) model (Absolute Value GARCH), the conditional variance is also exchanged by the conditional standard deviation. Additionally, since Taylor (1986) found that autocorrelation of absolute returns is often larger than autocorrelation of squared returns, lagged squared

errors are replaced by lagged absolute errors in the model equation, compared to the standard GARCH( $q, p$ ) model. A Box-Cox power transformation is applied to the ARCH and GARCH term in the APGARCH( $q, p$ ) model (Asymmetric Power GARCH), where the exact transformation is estimated based on the empirical data the model is applied to, in form of the power parameter  $\delta$ . The ARCH term is further split into a size effect component, in form of absolute lagged errors  $|\epsilon_{t-j}|$ , and a sign effect component  $\gamma_j \epsilon_{t-j}$ , which makes it possible to capture asymmetric volatility. In the NGARCH( $q, p$ ) model (Nonlinear GARCH), the standard GARCH( $q, p$ ) model is modified by also using lagged absolute errors instead of lagged squared errors, and a Box-Cox power transformation for the conditional standard deviation and lagged absolute errors, where  $\lambda$  is the power parameter that needs to be estimated. This transformation is usually used to linearize otherwise nonlinear models (Higgins and Bera, 1992). The ARCH term of the NAGARCH( $q, p$ ) model (Nonlinear Asymmetric GARCH) is extended, compared to the standard GARCH( $q, p$ ) model, by multiplying the lagged conditional variance with the absolute, by  $\eta_{2j}$  shifted, and standardized error  $z_{t-j}$ .  $\eta_{2j}$  is responsible to model the leverage effect in this model. The FGARCH( $q, p$ ) model (family GARCH) is an omnibus model, so a generalization of many symmetric and asymmetric GARCH models. All GARCH-type models considered in this study, except the CGARCH( $q, p$ ) model, are special cases of this very general GARCH specification. It allows for different power parameters for the conditional standard deviation ( $\lambda$ ) and for the transformed lagged errors ( $\delta$ ), as well as a shift and rotation of the news impact curve, so the curve which represents the impact of short-term shocks on proceeding volatility, captured by the ARCH term of the model equation. The last model considered in this study is the CGARCH( $q, p$ ) model (Component GARCH), where the conditional variance is decomposed into a permanent and transitory component.  $q_t$  represents the permanent component in form of a first order autoregressive process.  $\sigma_{t-j}^2 - q_{t-j}$  is the transitory component of the conditional variance, and  $\epsilon_{t-j}^2 - q_{t-j}$  captures the effect of short-term shocks on proceeding volatility (Ghalanos, 2023).

For the empirical estimation, I used  $q$  equal to 10 lags for the ARCH( $q$ ) model. For all other models, I used  $q$  and  $p$  equal to 1 for the ARCH and GARCH term respectively. I took the implementations of these GARCH specifications from the rugarch package in R (Galanos, 2024).

## 4.5 Innovation Distribution Assumptions

As innovation distribution assumptions  $d$ , I employed multiple standard distributions, which are commonly used for financial time series, as well as more sophisticated distributions. The standard distributions are the Normal distribution (NORM), the Students

t-distribution (STD), and the Generalized Error Distribution (GED). Compared to the Normal distribution, the Students t-distribution and the GED add the flexibility to model leptokurtic innovations. To additionally account for potential skewness in innovations  $\epsilon_t$ , I also incorporated skewed versions of all three distributions (SNORM, SSTD & SGED). Furthermore, I included Johnson's Reparametrized  $S_u$ -distribution (JSU), which is a four-parameter distribution with parameters for the mean, the standard deviation, skewness, and kurtosis. It is derived from the Normal distribution through a functional transformation that introduces skewness and kurtosis (Garcia-Jorcano and Novales, 2021). The most sophisticated distribution I used, is the Generalized Hyperbolic distribution (GHYP), which nests multiple other distributions, like the Students t-distribution and the Normal Inverse Gaussian distribution. It is more versatile in terms of modeling skewness and shape of the innovation distribution than standard distributions, but comes with the cost of having to optimize a more complicated function, which is computationally expensive and more prone to optimization problems, since large sections of the likelihood surface for certain combinations of parameters are relatively flat (Galanos, 2024). I implemented all distributions in their standardized form, so with unit variance, using the rugarch package in R (Galanos, 2024).

In addition to parametric distributions, I also incorporated an empirical distribution into the study by using empirical standardized quantiles of the residuals remaining after fitting the observations within the estimation window. To obtain parameter estimates  $\hat{\Psi}$  for that case, I used quasi-maximum likelihood estimation, which means that I used the Normal distribution for the likelihood optimization. This approach still leads to consistent estimation, even if the true distribution is not Normal. To estimate  $VarR_{T+1}^\alpha$  based on the empirical distribution, I used the  $\alpha$ -quantile of the standardized empirical distribution  $q_\alpha^{emp}$ , and located and scaled it by using the obtained estimates for  $\hat{\mu}$  and  $\hat{\sigma}_{T+1}$ . To estimate  $ES_{T+1}^\alpha$ , I calculated the empirical expected value of the standardized empirical distribution for all observations which are lower or equal to  $q_\alpha^{emp}$ , and located and scaled it by  $\hat{\mu}$  and  $\hat{\sigma}_{T+1}$ :

$$\begin{aligned} VarR_{T+1}^\alpha(\tilde{\Omega}_T) &= \hat{\mu} + \hat{\sigma}_{T+1}(\tilde{\Omega}_T)q_\alpha^{emp} \\ ES_{T+1}^\alpha(\tilde{\Omega}_T) &= \hat{\mu} + \hat{\sigma}_{T+1}(\tilde{\Omega}_T) \frac{1}{\sum_{t=1}^T \mathbb{I}(\epsilon_t \leq q_\alpha^{emp})} \sum_{t=1}^T \mathbb{I}(\epsilon_t \leq q_\alpha^{emp}) \epsilon_t \end{aligned} \quad (4.11)$$

With this broad set of distributions, I intended to explore whether more sophisticated distribution assumptions, capable of modeling skewness and shape more accurately, add significant value to financial risk forecasting compared to standard distributions, or if they primarily increase the number of parameters that must be estimated, which would con-

flict with the principle of model parsimony.

## 4.6 Empirical Modeling Approach

For the empirical study, I combined each considered GARCH specification  $V$  with each considered distribution assumption  $d$ , and calculated a series of historical one-step-ahead VaR and ES forecasts using a rolling-estimation window with a window length of  $T$ . This means that out of the whole series available, I initially selected the first  $T$  returns to estimate the VaR and ES for  $T + 1$ . Subsequently, the window was shifted one observation to the right, meaning that the second observation of the previous window became the first observation of the new window, and the last observation of the old window became the second-to-last observation of the new window. Based on this, the parameters were re-estimated, and the next one-step-ahead VaR and ES forecasts were calculated. After that, the window was shifted again. This process was repeated up to the last observation of the whole sample. Each resulting VaR and ES forecast was then compared with the actual historical return using the backtests and loss functions explained in the following subsections. I decided to use a window length of  $T = 750$  observations. Based on the findings of Righi and Ceretta (2015), a longer window length is particularly important for generating reliable ES forecasts. In their study, a window length of 1000 observations led to the best ES forecasts. On the other hand, Angelidis et al. (2004) found that shorter window lengths, around 250 to 500 observations, worked better for VaR forecasting. Considering these two perspectives, I chose 750 observations as a reasonable compromise. It provides enough data points for reliable VaR and ES estimation while leaving plenty of observations for forecasting and therefore backtesting. Additionally, it ensures that the estimation window still reflects relatively recent market dynamics. For the tolerance level  $\alpha$  I decided to choose 5%.

## 4.7 Backtesting

### 4.7.1 Value at Risk

To backtest VaR estimates, I used a conditional coverage test developed by Christoffersen (1998), which jointly tests, if the empirical proportion of VaR violations corresponds with the tolerance level  $\alpha$ , and if these VaR violations are independent over time. The test uses a sequence of indicator variables  $I_t$ , which represent the violation process of VaR estimates and are defined as follows:

$$I_t = \begin{cases} 1 & \text{if } r_t \leq VaR_t^\alpha \\ 0 & \text{if } r_t > VaR_t^\alpha \end{cases} \quad (4.12)$$

The null hypothesis of the Christofferson 2 test is that the expected value of  $I_t$ , conditioned on  $I_{t-1}$ , is equal to the tolerance level  $\alpha$ :

$$\begin{aligned} H_0: E[I_t | I_{t-1}] &= \alpha \\ H_1: E[I_t | I_{t-1}] &\neq \alpha \end{aligned} \quad (4.13)$$

To test this statistically, Christoffersen (1998) suggests a likelihood ratio test, which compares the restricted likelihood  $L_r$ , so the likelihood under  $H_0$ , against the unrestricted likelihood  $L_{ur}$ , which allows for different violation proportions depending on whether the previous observation  $I_{t-1}$  was a violation or not. With that, the test has power against models that lead to a wrong proportion of violations, as well as against models that lead to violation clustering. The resulting test statistic  $LR_{cc}$  is asymptotically Chi-squared distributed, with two degrees of freedom:

$$\begin{aligned} L_r(\alpha; I_1, I_2, \dots, I_T) &= \alpha^{n_1} (1 - \alpha)^{n_0} \\ L_{ur}(\hat{\Pi}; I_1, I_2, \dots, I_T) &= \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{11}^{n_{11}} (1 - \hat{\pi}_{11})^{n_{10}} \\ LR_{cc} &= -2 \log\left(\frac{L_r}{L_{ur}}\right) \stackrel{asy}{\sim} \chi^2(2) \end{aligned} \quad (4.14)$$

where  $n_{ij}$  is the number of observations with value  $i$  followed by  $j$ , where  $i$  and  $j$  are either 0 or 1. Additionally,  $\hat{\pi}_{i1} = \frac{n_{i1}}{n_{i0} + n_{i1}}$ .

#### 4.7.2 Expected Shortfall

For backtesting ES estimates, I used the conditional coverage test introduced by Du and Escanciano (2015). It is the analogous counterpart to the VaR backtest used in this paper, and has power against models which lead to wrong coverage of ES forecasts, and models which lead to serial dependence of errors in forecasting ES. Instead of using an indicator sequence which represents VaR violations with 1's and non-violations with 0's, the test is based on a sequence of cumulative violations  $H_t(\alpha)$ . Du and Escanciano (2015) have derived these cumulative violations based on the observation that the ES can be calculated by integrating VaR estimates from 0 to  $\alpha$  and dividing them by  $\alpha$  (see Equation 4.5). In their paper, they use the integral over the VaR violation process to construct  $H_t(\alpha)$ . Therefore, cumulative violations not only show if the VaR was exceeded, but also provide

information about the intensity of these violations. Because the whole lower tail of the distribution is incorporated in cumulative violations, and ES is a tail based risk measure, they provide an appropriate basis for testing ES estimates implicitly. They are defined as follows:

$$\begin{aligned} H_t(\alpha) &= \frac{1}{\alpha} \int_0^\alpha \mathbb{I}(u_t \leq u) du \\ &= \frac{1}{\alpha} (\alpha - u_t) \mathbb{I}(u_t \leq \alpha) \end{aligned} \quad (4.15)$$

where  $u_t = G(r_t, \Omega_{t-1})$ , with  $G$  being the cdf of the returns and  $\Omega_{t-1}$  the information set at time  $t - 1$ . As the distribution  $G$  is generally unknown, only estimated cumulative violations  $\hat{H}_t(\alpha)$  are feasible, calculated based on an distribution assumption  $d$ .

$$\begin{aligned} \hat{H}_t(\alpha) &= \frac{1}{\alpha} (\alpha - \hat{u}_t) \mathbb{I}(\hat{u}_t \leq \alpha) \\ \hat{u}_t &= D(r_t, \tilde{\Omega}_{t-1}, \hat{\theta}) \end{aligned} \quad (4.16)$$

In the empirical study,  $D$  is the cdf of the assumed distribution for the tested model, with  $\hat{\theta}$  summarizing all parameter estimates needed for this distribution, and estimated while fitting the model on preceding returns summarized in  $\tilde{\Omega}_{t-1}$ .

If the tested model correctly forecasts ES, the mean of  $\hat{H}_t(\alpha)$  should not be significantly different from  $\frac{\alpha}{2}$ . The corresponding hypothesis pair is:

$$\begin{aligned} H_0: E[H_t(\alpha) | \Omega_{t-1}] &= \frac{\alpha}{2} \\ H_1: E[H_t(\alpha) | \Omega_{t-1}] &\neq \frac{\alpha}{2} \end{aligned} \quad (4.17)$$

To test  $H_0$  statistically, Du and Escanciano (2015) propose a Portmanteau Box-Pierce test applied to the sequence of estimated cumulative violations  $\hat{H}_t(\alpha)$ . In their paper, they introduce a basic version of the test, which has a known distribution only if the in-sample period is significantly larger than the out-of-sample period, and a more general version, which explicitly respects parameter estimation uncertainty while calculating the test statistic, and can therefore also be used if the in-sample period is not significantly larger than the out-of-sample period. Since the in-sample period in this study is smaller than the out-of-sample period, I used the more general version of the test, which leads to a test statistic  $C_{ES}(m)$  that converges in distribution to a Chi-square distribution with  $m$  degrees of freedom, where  $m$  is the number of lags considered in the test:

$$\begin{aligned}
C_{ES}(m) &= n\hat{\rho}_m^\top \hat{\Sigma}^{-1} \hat{\rho}_m \\
C_{ES}(m) &\xrightarrow{d} \chi^2(m)
\end{aligned} \tag{4.18}$$

$\hat{\rho}_m$  is a vector of estimated correlation coefficients between  $\hat{H}_t(\alpha)$  and  $\hat{H}_{t-j}(\alpha)$  up to  $j = m$ .  $\hat{\Sigma}^{-1}$  is the inverse of  $\hat{\Sigma}$ , and is used to correct the test statistic for parameter estimation uncertainty. The  $i, j$ -th element of  $\hat{\Sigma}$  is  $\hat{\Sigma}_{ij} = \mathbb{I}(i = j) + \frac{n}{T} \hat{R}_i^\top W_T \hat{R}_j$ , with  $n$  being the number of forecasts and  $T$  the window length used to estimate the parameters.  $W_T$  is a consistent estimator of the covariance matrix of the models' parameter vector. In this study, I used the mean of parameter covariance estimators of all estimation windows. An exact expression for  $\hat{R}_j$  can be found in the footnote<sup>1</sup>. I used  $m = 5$  lags, as suggested by Du and Escanciano (2015).

## 4.8 Loss Function

As the aforementioned backtests can only be used to statistically validate models by providing information on whether a model significantly violates its assumptions, it cannot be used to rank models. This is insufficient for determining the best-performing model. Loss functions, on the other hand, measure the loss associated with using a model and can therefore be used to rank models based on their performance. To work out the best-performing models of this study, I used the FZ loss function from Fissler and Ziegel (2016). It is a scoring function which calculates the loss of VaR and ES estimates jointly. In their paper, Fissler and Ziegel (2016) introduce a general loss function and theoretically show that the true VaR and ES forecasts uniquely minimize the expected score. This is because the pair of VaR and ES are elicitable under mild conditions, meaning there exists a loss function for which the true forecast is the unique minimizer of the expected value of that loss function. I used a specific version of this loss function, as defined in Patton et al. (2019), to calculate the loss associated with each considered model. It requires that  $Var_t^\alpha < 0$  and  $ES_t^\alpha < 0$ , which can generally be expected. It is defined as follows:

$$FZ_t^\alpha(Var_t^\alpha, ES_t^\alpha, r_t) = \frac{1}{\alpha ES_t^\alpha} \mathbb{I}(r_t \leq Var_t^\alpha) (Var_t^\alpha - r_t) + \frac{Var_t^\alpha}{ES_t^\alpha} + \log(-ES_t^\alpha) - 1 \tag{4.19}$$

It penalizes strong violations in proportion to VaR and ES estimates in cases where the VaR is exceeded, but also high estimates of both risk measures, which is especially important in cases where the VaR is not exceeded.

<sup>1</sup>  $\hat{R}_j = \frac{1}{\alpha(\frac{1}{3} - \frac{\alpha}{4})} \frac{1}{n-j} \sum_{t=j+1}^n (\hat{H}_{t-j}(\alpha) - \frac{\alpha}{2}) g_\epsilon(\hat{\epsilon}_t) \mathbb{I}(\hat{\epsilon}_t \leq G_\epsilon^{-1}(\alpha)) \frac{\dot{\mu}_t + \hat{\epsilon}_t \dot{\sigma}_t}{\dot{\sigma}_t}$  with  $\dot{\mu}_t = \frac{\partial \mu(\bar{\Omega}_{t-1}, \hat{\theta})}{\partial \theta}$ ,  $\dot{\sigma}_t = \frac{\partial \sigma(\bar{\Omega}_{t-1}, \hat{\theta})}{\partial \theta}$  and  $g_\epsilon$  being the pdf of the assumed innovation distribution with expected value of 0.



To test for statistical significant differences in the loss associated with two competing models, and with that, to determine whether one model significantly outperforms the other in terms of FZ Loss by having a lower mean FZ loss, I used the DM test developed by Diebold and Mariano (2002). It is a t-test which evaluates whether the expected value of the differential series  $d_{ij}$ , so the difference between the two loss sequences of the two compared models, is significantly different from zero ( $H_0 : E[d_{ij}] = 0$ ;  $H_1 : E[d_{ij}] \neq 0$ ). If that is the case, it can be deduced that one model is significantly better than the other. For that, the following test statistic is used:

$$DM_{ij} = \frac{\bar{d}_{ij}}{\hat{\sigma}_{\bar{d}_{ij}}} \quad (4.20)$$

$$DM_{ij} \stackrel{asy}{\sim} N(0, 1)$$

where  $\bar{d}_{ij}$  is the mean of the empirical differential sequence, with  $\hat{\sigma}_{\bar{d}_{ij}}$  being its consistent standard deviation estimator. Like Patton et al. (2019), I also used Newey-West standard errors to calculate  $\hat{\sigma}_{\bar{d}_{ij}}$ .

## 4.9 Evaluation Approach and Ranking

To finally come to a statement about the best-performing models, I brought all previously explained steps, the two backtests and the loss function, together into a three-step decision process. For a model to be considered in the ranking based on the FZ loss function, it first has to pass both backtests. This is to make sure that no assumption is significantly violated (violation proportion of  $\alpha$  and no serial dependence of violations), and the model is valid in terms of these assumptions. I set the boundary for passing to a  $p$ -value of 0.1. All models with higher  $p$ -values pass the tests, while the rest do not pass. After that, I calculated the mean of the FZ loss sequences for all models which have passed the backtesting. Based on this, I ranked these models within the corresponding index in ascending order, meaning that the model with the lowest mean FZ loss has rank 1. Afterwards, I compared each subsequent model with the model with rank 1, using the DM test. If the resulting  $p$ -value is greater than 0.1, I consider the model to be part of the set of best-performing models.

The test results are presented in the next section in form of tables, where the column names are all examined distributions, and the row names are all examined GARCH specifications. The first two tables of each index present the results of the VaR and ES backtests, respectively. When I use the term "model," I am referring to a specific combination of a GARCH specification and a distribution assumption. When I mention the name of

a GARCH specification or distribution, I am referring to all models that fall under that specification or distribution. For the conditional coverage tests of VaR and ES estimates, I report the  $p$ -values.  $p$ -values above 0.1 are marked in green. I consider these models to be valid in terms of correct violation proportions and serial independent violations. In cases where the  $p$ -value is between 0.05 and 0.1, it is marked in orange. I consider the validity of these models as doubtful, and consequently do not consider them in the ranking stage. If the  $p$ -values is equal or lower than 0.05, I reject the null of a valid model for the corresponding index, and mark the them in red.

The third table of each index presents the results of the ranking based on mean FZ loss. Models which have not passed to this stage hold a red "np" in their cell. For the rest, the corresponding rank is reported either in green, if the mean FZ loss of the corresponding model is not significantly different from the model with rank 1 ( $p > 0.1$ ), or in orange, in cases were a significant difference in mean FZ loss can be found ( $p \leq 0.1$ ). I consider all models which have a green rank to be part of the set of best-performing models for the corresponding index. The top 10 best-performing models of each index are additionally marked with an asterisk (\*), and reported in a separate table at the end of the conclusion section.

## 5 Empirical Results

### 5.1 DAX

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.452 | 0.014 | 0.001 | 0.653 | 0.317 | 0.186 | 0.677 | 0.526 | 0.926     |
| GARCH     | 0.483 | 0.013 | 0.111 | 0.918 | 0.381 | 0.918 | 0.809 | 0.677 | 0.519     |
| EGARCH    | 0.257 | 0.020 | 0.111 | 0.859 | 0.483 | 0.993 | 0.672 | 0.653 | 0.918     |
| GJR-GARCH | 0.536 | 0.005 | 0.177 | 0.683 | 0.583 | 0.630 | 0.993 | 0.859 | 0.797     |
| TGARCH    | 0.452 | 0.085 | 0.219 | 0.916 | 0.672 | 0.818 | 0.669 | 0.672 | 0.818     |
| AVGARCH   | 0.536 | 0.141 | 0.294 | 0.902 | 0.859 | 0.902 | 0.916 | 0.623 | 0.818     |
| APARCH    | 0.526 | 0.125 | 0.162 | 0.918 | 0.859 | 0.780 | 0.916 | 0.677 | 0.780     |
| NGARCH    | 0.483 | 0.007 | 0.205 | 0.868 | 0.526 | 0.896 | 0.526 | 0.381 | 0.683     |
| NAGARCH   | 0.317 | 0.013 | 0.063 | 0.902 | 0.623 | 0.967 | 0.942 | 0.672 | 0.993     |
| FGARCH    | 0.381 | 0.013 | 0.063 | 0.902 | 0.809 | 0.990 | 0.868 | 0.809 | 0.967     |
| CGARCH    | 0.315 | 0.002 | 0.111 | 0.747 | 0.177 | 0.809 | 0.427 | 0.266 | 0.916     |

**Table 5.1: DAX: VaR Conditional Coverage Backtest  $p$ -Values**

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.332 | 0.457 | 0.002 | 0.223 | 0.413 | 0.000 | 0.398 | 0.440 | 0.070     |
| GARCH     | 0.061 | 0.211 | 0.221 | 0.093 | 0.194 | 0.172 | 0.177 | 0.191 | 0.020     |
| EGARCH    | 0.148 | 0.144 | 0.163 | 0.188 | 0.153 | 0.160 | 0.159 | 0.153 | 0.089     |
| GJR-GARCH | 0.575 | 0.728 | 0.761 | 0.857 | 0.706 | 0.830 | 0.751 | 0.744 | 0.528     |
| TGARCH    | 0.605 | 0.488 | 0.558 | 0.582 | 0.497 | 0.539 | 0.486 | 0.497 | 0.445     |
| AVGARCH   | 0.576 | 0.547 | 0.663 | 0.579 | 0.416 | 0.563 | 0.502 | 0.711 | 0.299     |
| APARCH    | 0.806 | 0.727 | 0.786 | 0.775 | 0.701 | 0.769 | 0.718 | 0.652 | 0.671     |
| NGARCH    | 0.200 | 0.318 | 0.566 | 0.167 | 0.475 | 0.326 | 0.492 | 0.657 | 0.193     |
| NAGARCH   | 0.503 | 0.375 | 0.564 | 0.624 | 0.591 | 0.567 | 0.575 | 0.585 | 0.316     |
| FGARCH    | 0.841 | 0.449 | 0.776 | 0.696 | 0.475 | 0.706 | 0.720 | 0.400 | 0.652     |
| CGARCH    | 0.094 | 0.094 | 0.109 | 0.098 | 0.144 | 0.135 | 0.227 | 0.152 | 0.054     |

Table 5.2: DAX: ES Conditional Coverage Backtest  $p$ -Values

|           | NORM | STD | GED | SNORM | SSTD | SGED | GHYP | JSU | EMPIRICAL |
|-----------|------|-----|-----|-------|------|------|------|-----|-----------|
| ARCH      | 74   | np  | np  | 68    | 42   | np   | 44   | 40  | np        |
| GARCH     | np   | np  | 69  | np    | 54   | 64   | 58   | 53  | np        |
| EGARCH    | 76   | np  | 73  | 67    | 59   | 51   | 60   | 57  | np        |
| GJR-GARCH | 55   | np  | 31  | 27    | 7*   | 3*   | 10*  | 8*  | 35        |
| TGARCH    | 43   | np  | 33  | 32    | 12   | 11   | 14   | 13  | 16        |
| AVGARCH   | 77   | 49  | 29  | 25    | 18   | 1*   | 28   | 78  | 71        |
| APARCH    | 65   | 41  | 30  | 17    | 19   | 15   | 21   | 23  | 36        |
| NGARCH    | 75   | np  | 66  | 70    | 52   | 48   | 46   | 45  | 62        |
| NAGARCH   | 50   | np  | np  | 20    | 5*   | 2*   | 6*   | 4*  | 22        |
| FGARCH    | 37   | np  | np  | 38    | 26   | 34   | 24   | 39  | 9*        |
| CGARCH    | np   | np  | 72  | np    | 61   | 63   | 47   | 56  | np        |

Table 5.3: DAX: Ranks of Models based on FZ Loss

In Table 5.1, it can be seen that models using the Students t-distribution mostly failed the VaR Christofferson 2 backtest. Only in combination with the AVGARCH and APARCH specification, the Students t-distribution passed. Additionally, also the combination of GED with the ARCH, NAGARCH, and FGARCH specifications did not pass. All other models have a  $p$ -value above 0.1, meaning they can not be rejected. Similarly, also the ES conditional coverage backtest was passed by most of the models. Only a few ARCH, GARCH, and CGARCH models have a  $p$ -value below 0.1, and therefore did not proceed to the ranking stage. The empirical distribution has, with four, the highest number of models that did not pass the ES backtest among all distributions.

The ranking results in Table 5.3 strengthen the picture that many models did not perform significantly different from each other. However, a certain pattern can be observed. No EGARCH model is part of the set of best-performing models. Also the ARCH, GARCH, and CGARCH models, all symmetric specifications, have comparably less models in the set of best-performing models. When they are included, they still tend to hold higher ranks. Specifications which hold the lowest ranks are the GJR-GARCH, TGARCH, and NAGARCH specifications, combined with the skewed Student's  $t$ -, skewed GED, Generalized Hyperbolic, and Johnson's Reparametrized  $S_u$ -distribution. All three GARCH specifications have in common that they have a parameter in the ARCH term that can model asymmetric volatility. The distributions of the models with the lowest ranks all have the capability to model skewness and kurtosis in innovations. The AVGARCH specification combined with the skewed GED hold rank 1, which is somehow striking, because other than the previously mentioned specifications, it is a symmetric specification, and all other models of this specification have a notably higher rank. Looking at distributions, the skewed GED has the models with the three lowest ranks.

To summarize the results, most models led to valid results in terms of conditional violation proportion for VaR and ES estimation and could therefore be used. Standard specifications, combined with a distribution which allows for skewness and kurtosis, do not perform significantly worse than more complex models. This could be because many complex specifications include standard specifications as special cases, so the estimated parameters might end up in a way that the resulting model resembles the standard specifications. That said, if the goal is to use the absolute best-performing models, GARCH specifications, capable of modeling asymmetric volatility in combination with distributions allowing for skewness and kurtosis in innovations, should be preferred. Models using the GJR-GARCH and NAGARCH specification, combined with the skewed Student's  $t$ -, GED, or Johnson's Reparametrized  $S_u$ -distribution, performed particularly well. The Generalized Hyperbolic distribution, when combined with these two specifications, also performed very well, but it comes with the cost of a more complicated function with additional parameters, which have to be estimated. For this reason, the other three mentioned distributions should be preferred. The results indicate that the GARCH specification of a model seems to be the stronger driver for its performance, while the distribution properties still play an important role to find the best model for DAX.

## 5.2 WIG

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.007 | 0.093 | 0.000 | 0.315 | 0.143 | 0.000 | 0.342 | 0.365 | 0.365     |
| GARCH     | 0.637 | 0.197 | 0.637 | 0.605 | 0.205 | 0.894 | 0.392 | 0.203 | 0.386     |
| EGARCH    | 0.371 | 0.391 | 0.350 | 0.187 | 0.323 | 0.154 | 0.291 | 0.257 | 0.342     |
| GJR-GARCH | 0.902 | 0.315 | 0.891 | 0.825 | 0.902 | 0.825 | 0.891 | 0.891 | 0.891     |
| TGARCH    | 0.323 | 0.580 | 0.647 | 0.239 | 0.632 | 0.191 | 0.519 | 0.519 | 0.392     |
| AVGARCH   | 0.122 | 0.487 | 0.257 | 0.086 | 0.257 | 0.154 | 0.257 | 0.257 | 0.205     |
| APARCH    | 0.868 | 0.863 | 0.768 | 0.488 | 0.983 | 0.577 | 0.699 | 0.768 | 0.965     |
| NGARCH    | 0.365 | 0.391 | 0.391 | 0.868 | 0.350 | 0.408 | 0.291 | 0.350 | 0.381     |
| NAGARCH   | 0.391 | 0.381 | 0.647 | 0.257 | 0.605 | 0.349 | 0.519 | 0.566 | 0.391     |
| FGARCH    | 0.537 | 0.342 | 0.637 | 0.188 | 0.386 | 0.623 | 0.647 | 0.868 | 0.637     |
| CGARCH    | 0.365 | 0.093 | 0.381 | 0.768 | 0.580 | 0.566 | 0.637 | 0.637 | 0.203     |

Table 5.4: WIG: VaR Conditional Coverage Backtest  $p$ -Values

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.000 | 0.002 | 0.000 | 0.000 | 0.001 | 0.000 | 0.001 | 0.001 | 0.000     |
| GARCH     | 0.012 | 0.008 | 0.002 | 0.001 | 0.003 | 0.000 | 0.001 | 0.001 | 0.011     |
| EGARCH    | 0.008 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.008     |
| GJR-GARCH | 0.067 | 0.139 | 0.054 | 0.024 | 0.082 | 0.015 | 0.053 | 0.057 | 0.088     |
| TGARCH    | 0.001 | 0.003 | 0.001 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | 0.001     |
| AVGARCH   | 0.394 | 0.016 | 0.016 | 0.000 | 0.006 | 0.004 | 0.008 | 0.003 | 0.370     |
| APARCH    | 0.324 | 0.331 | 0.231 | 0.142 | 0.222 | 0.098 | 0.171 | 0.217 | 0.327     |
| NGARCH    | 0.117 | 0.179 | 0.076 | 0.022 | 0.091 | 0.026 | 0.060 | 0.067 | 0.106     |
| NAGARCH   | 0.004 | 0.008 | 0.006 | 0.002 | 0.002 | 0.001 | 0.002 | 0.001 | 0.007     |
| FGARCH    | 0.328 | 0.042 | 0.032 | 0.001 | 0.404 | 0.067 | 0.203 | 0.027 | 0.348     |
| CGARCH    | 0.009 | 0.007 | 0.004 | 0.002 | 0.004 | 0.001 | 0.002 | 0.001 | 0.009     |

Table 5.5: WIG: ES Conditional Coverage Backtest  $p$ -Values

|           | NORM | STD | GED | SNORM | SSTD | SGED | GHYP | JSU | EMPIRICAL |
|-----------|------|-----|-----|-------|------|------|------|-----|-----------|
| ARCH      | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| GARCH     | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| EGARCH    | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| GJR-GARCH | np   | 12  | np  | np    | np   | np   | np   | np  | np        |
| TGARCH    | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| AVGARCH   | 14   | np  | np  | np    | np   | np   | np   | np  | 13        |
| APARCH    | 10*  | 5*  | 7*  | 9*    | 2*   | np   | 4*   | 3*  | 8*        |
| NGARCH    | 17   | 15  | np  | np    | np   | np   | np   | np  | 16        |
| NAGARCH   | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| FGARCH    | 18   | np  | np  | np    | 6*   | np   | 1*   | np  | 11        |
| CGARCH    | np   | np  | np  | np    | np   | np   | np   | np  | np        |

Table 5.6: WIG: Ranks of Models based on FZ Loss

The results for WIG show stronger differences in model performance than the results for DAX. While the Christofferson 2 backtest results in Table 5.4 only show VaR model rejections of some ARCH models, way more models can be rejected for ES estimation (Table 5.5). The models that mainly passed the ES backtest are those capable of modeling asymmetric volatility and having a power parameter. One of them is the NGARCH model, which has a power parameter but cannot model asymmetric volatility. Most of the  $p$ -values for this specification are around 0.1, meaning that only some distribution assumptions in combination with the NGARCH specification passed. The APARCH specification, with its power parameter and ability to model asymmetric volatility, performed particularly well. Only one distribution assumption, namely the skewed GED, combined with the APARCH specification, led to a  $p$ -value below 0.1, while all other distributions passed the ES backtest. Additionally, the FGARCH specification, when combined with the Normal, skewed Student's  $t$ -, Generalized Hyperbolic, and empirical distribution, passed the ES backtest. This specification is also capable of modeling asymmetric volatility and it furthermore incorporates two different power parameters. Interestingly, also the AVGARCH specification in combination with the Normal and empirical distribution passed the backtest, which is somehow surprising, because in contrast to the other specifications that passed the backtest, it is not capable of modeling asymmetric volatility, has no power parameter, and the combination with the Normal distribution does not even allow for modeling skewness and kurtosis in innovations.

Looking at the model rankings in Table 5.6, it can be seen that most of the lowest ranks are held by APARCH models. All APARCH models, apart from the one with a skewed GED distribution, are among the top 10 best-performing models. The model with rank

1 is the FGARCH specification combined with the Generalized Hyperbolic distribution, representing the combination of the most versatile GARCH specification with the most sophisticated distribution. Moreover, all APARCH, FGARCH, and AVGARCH models that proceeded to the ranking stage, have a mean FZ loss that is not significantly higher than the one of the FGARCH - Generalized Hyperbolic distribution combination, meaning they are also part of the set of best-performing models for WIG.

This results indicates that for ES estimation of WIG, the GARCH specification plays a more important role than the distribution assumption. Specifications which allow for asymmetric volatility, and have a power parameter, are the most suitable ones, with APARCH being the overall best-performing specification for WIG in this study. Within this specification, distributions which allow for skewness and excess kurtosis have lower ranks than the ones without these features, but the differences are not significant. However, if the goal is to only estimate the VaR for WIG without ES estimation, more standard models that incorporate an ARCH and GARCH term might be sufficient. This claim would have to be further tested with a loss function designed to solely rank VaR models. If standard models perform at least equally as well, they should be preferred, as more sophisticated models come with additional parameters that need to be estimated, leading to more demanding computations, a higher risk of estimation issues (e.g., convergence problems), and they conflict with the spirit of model parsimony.

### 5.3 Bitcoin

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.050 | 0.413 | NA    | 0.011 | 0.251 | NA    | 0.268 | 0.182 | 0.340     |
| GARCH     | 0.025 | 0.250 | 0.171 | 0.002 | 0.069 | 0.268 | 0.381 | 0.182 | 0.381     |
| EGARCH    | 0.002 | 0.824 | 0.111 | 0.001 | 0.412 | 0.108 | 0.439 | 0.534 | 0.116     |
| GJR-GARCH | 0.055 | 0.221 | 0.159 | 0.007 | 0.071 | 0.198 | 0.541 | 0.127 | 0.604     |
| TGARCH    | 0.001 | 0.906 | 0.000 | 0.000 | 0.396 | 0.000 | 0.554 | 0.656 | 0.175     |
| AVGARCH   | 0.018 | 0.585 | 0.000 | 0.002 | 0.139 | 0.000 | 0.495 | 0.534 | 0.173     |
| APARCH    | 0.022 | 0.790 | 0.171 | 0.022 | 0.243 | 0.258 | 0.441 | 0.463 | 0.375     |
| NGARCH    | 0.070 | 0.827 | 0.171 | 0.007 | 0.382 | 0.207 | 0.319 | 0.259 | 0.439     |
| NAGARCH   | 0.087 | 0.350 | 0.138 | 0.033 | 0.136 | 0.222 | 0.182 | 0.197 | 0.564     |
| FGARCH    | 0.138 | 0.139 | 0.507 | 0.111 | 0.066 | 0.712 | 0.132 | 0.224 | 0.375     |
| CGARCH    | 0.043 | 0.088 | 0.207 | 0.007 | 0.115 | 0.365 | 0.820 | 0.241 | 0.495     |

**Table 5.7: Bitcoin: VaR Conditional Coverage Backtest  $p$ -Values**

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.122 | 0.318 | 0.000 | 0.000 | 0.299 | 0.000 | 0.342 | 0.346 | 0.043     |
| GARCH     | 0.483 | 0.257 | 0.250 | 0.389 | 0.269 | 0.388 | 0.301 | 0.314 | 0.429     |
| EGARCH    | 0.337 | 0.318 | 0.220 | 0.311 | 0.239 | 0.261 | 0.217 | 0.231 | 0.299     |
| GJR-GARCH | 0.726 | 0.201 | 0.229 | 0.639 | 0.099 | 0.178 | 0.128 | 0.089 | 0.401     |
| TGARCH    | 0.237 | 0.370 | 0.000 | 0.297 | 0.310 | 0.000 | 0.247 | 0.263 | 0.152     |
| AVGARCH   | 0.293 | 0.241 | 0.001 | 0.205 | 0.249 | 0.010 | 0.068 | 0.110 | 0.199     |
| APARCH    | 0.308 | 0.290 | 0.240 | 0.393 | 0.174 | 0.212 | 0.179 | 0.225 | 0.170     |
| NGARCH    | 0.328 | 0.251 | 0.248 | 0.202 | 0.232 | 0.228 | 0.223 | 0.249 | 0.357     |
| NAGARCH   | 0.627 | 0.198 | 0.225 | 0.656 | 0.141 | 0.310 | 0.219 | 0.155 | 0.490     |
| FGARCH    | 0.819 | 0.626 | 0.630 | 0.422 | 0.047 | 0.197 | 0.340 | 0.627 | 0.675     |
| CGARCH    | 0.414 | 0.237 | 0.338 | 0.373 | 0.194 | 0.322 | 0.272 | 0.286 | 0.449     |

Table 5.8: Bitcoin: ES Conditional Coverage Backtest  $p$ -Values

|           | NORM | STD | GED | SNORM | SSTD | SGED | GHYP | JSU | EMPIRICAL |
|-----------|------|-----|-----|-------|------|------|------|-----|-----------|
| ARCH      | np   | 59  | np  | np    | 55   | np   | 60   | 54  | np        |
| GARCH     | np   | 29  | 19  | np    | np   | 35   | 32   | 36  | 39        |
| EGARCH    | np   | 31  | 1*  | np    | 47   | 9*   | 20   | 37  | 25        |
| GJR-GARCH | np   | 48  | 10* | np    | np   | 41   | 51   | np  | 44        |
| TGARCH    | np   | 6*  | np  | np    | 28   | np   | 16   | 22  | 34        |
| AVGARCH   | np   | 21  | np  | np    | 42   | np   | np   | 46  | 62        |
| APARCH    | np   | 33  | 5*  | np    | 43   | 18   | 27   | 30  | 56        |
| NGARCH    | np   | 4*  | 8*  | np    | 14   | 13   | 11   | 12  | 63        |
| NAGARCH   | np   | 57  | 23  | np    | 61   | 45   | 58   | 50  | 17        |
| FGARCH    | 66   | 49  | 24  | 65    | np   | 52   | 40   | 53  | 64        |
| CGARCH    | np   | np  | 3*  | np    | 26   | 2*   | 7*   | 15  | 38        |

Table 5.9: Bitcoin: Ranks of Models based on FZ Loss

When looking at the VaR backtesting results for Bitcoin in Table 5.7, it can be seen that especially the Normal distribution, both in its symmetric and skewed form, did not perform well in VaR forecasting. Most of the  $p$ -values are below 0.1, and therefore did not pass this backtest. Only in combination with the FGARCH specification, both distributions passed with a  $p$ -value slightly above 0.1. Additionally, models composed of the TGARCH and AVGARCH specification, with the symmetric and skewed GED, also have to be rejected with a  $p$ -value below 0.05. Models using a skewed Students t-distribution along with the GARCH, GJR-GARCH, and FGARCH specification, and the combination of the CGARCH specification and the Students t-distribution, failed to exceed a  $p$ -value of 0.1. All other models passed the VaR backtest. Turning to the ES backtest results in Table 5.8,



it can be seen that this backtest was passed by most of the models, and a clear pattern is hardly observable. Yet, as it was the case in the VaR backtests, the TGARCH and AVGARCH specification, when paired with the symmetric and skewed GED, failed the ES backtest, with  $p$ -values close to 0. Also, the ARCH specification, combined with the GED, skewed Normal, skewed GED, and empirical distribution, ended up with a  $p$ -value below 0.05. The GJR-GARCH specification, along with the skewed Students  $t$ - and Johnson's Reparametrized  $S_u$ -distribution, the AVGARCH specification combined with the Generalized Hyperbolic distribution, and the FGARCH specification together with the skewed Students  $t$ -distribution, failed to proceed to the ranking stage as well.

The ranking results in Table 5.9 further support the observation from the backtests that, for Bitcoin, many models performed similarly well, without significant differences in ranking results. Accordingly, the set of best-performing models is relatively large. In terms of distributions, particularly many models using the Students  $t$ -, GED, and Johnson's Reparametrized  $S_u$ -distribution, are part of this set. Seven out of the top 10 best-performing models based on FZ loss use either the Students  $t$ -distribution or the GED, two symmetric distributions. This shows a difference of Bitcoin results from the results of the other indices, as in the case of the other indices, models with skewed distributions usually occupy the lowest ranks. The Normal distribution in its symmetric and skewed form, on the other hand, performed the worst out of all distributions for Bitcoin. From the GARCH specifications, especially the NGARCH and CGARCH performed well, with most of the models using this specification having comparably low ranks. The models with rank 1 is the EGARCH model with the GED.

Overall, it seems that for Bitcoin the innovation distribution assumption plays a more important role to find the best-performing models than the GARCH specification. The fact that the Normal and skewed Normal distribution were mostly rejected, while the other symmetrical distributions, namely the Students  $t$ -distribution and the GED, were the best-performing distributions, are an indicator that the most important feature of a distribution for Bitcoin is its ability to model excess kurtosis in innovations. The backtest results show that in the sample I used, the distribution assumption played a more pivotal role for VaR estimation than for ES estimation. In terms of the GARCH specifications, no clear pattern is observable. While especially NGARCH and CGARCH led to the lowest ranks, and therefore performed comparably the best, the difference to other specifications does not seem to be strong, since it is often not significant. The argument I have already used for the DAX results, that standard specifications are special cases of nearly all more complex specifications used in this study, and therefore the parameters of the complex specifications might be estimated in a way that the resulting model resembles the standard models, can also be used for Bitcoin. Nevertheless, the power parameter

in the NGARCH specification and the decomposition of volatility into a permanent and transitory component in the CGARCH specification still seem to add a small advantage. But this advantage comes with the price of more parameters that have to be estimated.

## 5.4 Gold

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.036 | 0.002 | 0.000 | 0.110 | 0.073 | 0.000 | 0.102 | 0.083 | 0.124     |
| GARCH     | 0.257 | 0.047 | 0.101 | 0.423 | 0.186 | 0.423 | 0.397 | 0.331 | 0.440     |
| EGARCH    | 0.117 | 0.015 | 0.083 | 0.103 | 0.117 | 0.236 | 0.117 | 0.117 | 0.139     |
| GJR-GARCH | 0.093 | 0.007 | 0.063 | 0.459 | 0.294 | 0.458 | 0.442 | 0.397 | 0.356     |
| TGARCH    | 0.117 | 0.012 | 0.073 | 0.111 | 0.102 | 0.236 | 0.246 | 0.117 | 0.097     |
| AVGARCH   | 0.019 | 0.004 | 0.029 | 0.111 | 0.093 | 0.117 | 0.111 | 0.102 | 0.024     |
| APARCH    | 0.135 | 0.006 | 0.065 | 0.455 | 0.126 | 0.423 | 0.397 | 0.115 | 0.454     |
| NGARCH    | 0.230 | 0.013 | 0.079 | 0.459 | 0.154 | 0.443 | 0.366 | 0.220 | 0.440     |
| NAGARCH   | 0.073 | 0.003 | 0.080 | 0.459 | 0.257 | 0.459 | 0.423 | 0.294 | 0.390     |
| FGARCH    | 0.154 | 0.004 | 0.036 | 0.331 | 0.196 | 0.397 | 0.366 | 0.135 | 0.440     |
| CGARCH    | 0.186 | 0.026 | 0.101 | 0.366 | 0.186 | 0.423 | 0.397 | 0.186 | 0.390     |

**Table 5.10: Gold: VaR Conditional Coverage Backtest  $p$ -Values**

|           | NORM  | STD   | GED   | SNORM | SSTD  | SGED  | GHYP  | JSU   | EMPIRICAL |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|
| ARCH      | 0.223 | 0.415 | 0.237 | 0.257 | 0.368 | 0.011 | 0.430 | 0.391 | 0.249     |
| GARCH     | 0.501 | 0.470 | 0.444 | 0.434 | 0.327 | 0.344 | 0.299 | 0.310 | 0.490     |
| EGARCH    | 0.027 | 0.048 | 0.034 | 0.029 | 0.038 | 0.032 | 0.029 | 0.033 | 0.028     |
| GJR-GARCH | 0.143 | 0.088 | 0.106 | 0.091 | 0.085 | 0.076 | 0.073 | 0.075 | 0.088     |
| TGARCH    | 0.023 | 0.051 | 0.033 | 0.027 | 0.047 | 0.034 | 0.035 | 0.042 | 0.033     |
| AVGARCH   | 0.022 | 0.806 | 0.674 | 0.065 | 0.103 | 0.948 | 0.042 | 0.094 | 0.057     |
| APARCH    | 0.484 | 0.677 | 0.292 | 0.213 | 0.279 | 0.218 | 0.218 | 0.254 | 0.308     |
| NGARCH    | 0.672 | 0.614 | 0.646 | 0.576 | 0.645 | 0.510 | 0.477 | 0.558 | 0.697     |
| NAGARCH   | 0.079 | 0.102 | 0.090 | 0.091 | 0.087 | 0.066 | 0.071 | 0.047 | 0.049     |
| FGARCH    | 0.201 | 0.731 | 0.145 | 0.145 | 0.166 | 0.056 | 0.116 | 0.101 | 0.192     |
| CGARCH    | 0.538 | 0.476 | 0.560 | 0.524 | 0.341 | 0.424 | 0.339 | 0.370 | 0.449     |

**Table 5.11: Gold: ES Conditional Coverage Backtest  $p$ -Values**

|           | NORM | STD | GED | SNORM | SSTD | SGED | GHYP | JSU | EMPIRICAL |
|-----------|------|-----|-----|-------|------|------|------|-----|-----------|
| ARCH      | np   | np  | np  | 40    | np   | np   | 37   | np  | 39        |
| GARCH     | 34   | np  | 26  | 28    | 23   | 15   | 17   | 20  | 27        |
| EGARCH    | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| GJR-GARCH | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| TGARCH    | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| AVGARCH   | np   | np  | np  | np    | np   | 25   | np   | np  | np        |
| APARCH    | 36   | np  | np  | 29    | 22   | 19   | 9*   | 14  | 31        |
| NGARCH    | 32   | np  | np  | 21    | 12   | 5*   | 7*   | 10* | 18        |
| NAGARCH   | np   | np  | np  | np    | np   | np   | np   | np  | np        |
| FGARCH    | 38   | np  | np  | 35    | 3*   | np   | 6*   | 30  | 33        |
| CGARCH    | 24   | np  | 16  | 13    | 11   | 1*   | 2*   | 8   | 4*        |

Table 5.12: Gold: Ranks of Models based on FZ Loss

In the case of Gold, many models with a symmetric innovation distribution assumption did not pass the VaR Christofferson 2 backtest, as presented in Table 5.10. All model with a Students t-distribution have a  $p$ -value below 0.05 and consequently have to be rejected. Additionally, most models using the GED have a  $p$ -value smaller than 0.1 and were therefore not considered in the ranking stage. For the Normal distribution, half of the models passed the VaR backtesting. From the perspective of GARCH specifications, the ARCH and AVGARCH models performed worse compared to the other specifications, even when combined with distributions that allow for skewness. While the distribution assumption had a greater influence on the VaR backtest results than the GARCH specification, the situation appears to be the opposite for the ES conditional coverage backtest results presented in Table 5.11. Especially the EGARCH and TGARCH specification did not perform well. All EGARCH models and all but one TGARCH model have to be rejected with a  $p$ -value below 0.05. Many models that use the GJR-GARCH and NAGARCH specification ended up with a  $p$ -value below 0.1 and hence did not proceed to the ranking stage. The AVGARCH models also performed relatively poorly, with only four models passing the ES backtest. In terms of distributions, the skewed GED performed the worst in the ES backtest, with the fewest models using this distribution passing that stage.

The ranking is dominated by models with the CGARCH, FGARCH and NGARCH specifications. Additionally, some APARCH models have ranks that are higher, though not significantly higher, than the model with rank 1. The FGARCH, NGARCH, and APARCH specifications share the characteristic of including a power parameter, which must be estimated when fitting the model to the estimation sample. It is notable that only one model with a symmetric distribution is part of the set of best-performing models, namely the CGARCH

- GED model. The lowest ranks are held by models with the skewed GED and Generalized Hyperbolic distribution. The combination of the CGARCH specification and skewed GED holds rank 1.

In summary, the results imply that for VaR estimation of Gold, the distribution assumption plays a more important role than the GARCH specification. Distributions that are capable of modeling skewness should be preferred. Conversely, for valid ES estimation, the GARCH specification seems to have a greater impact. Specifications that include a power parameter, standard specifications, and the CGARCH specification, with its decomposition into a permanent and transitory volatility component, led to more valid ES estimates than specifications which only add asymmetry to the standard GARCH model. The overall best specifications for joint VaR and ES estimation based on FZ loss are the CGARCH, FGARCH, and NGARCH, paired with distributions that allow for skewness and kurtosis, particularly the skewed GED and Generalized Hyperbolic distribution. For Gold both, the GARCH specification and the distribution assumption play a pivotal role in determining the best-performing model, and it cannot be said that one is more important than the other.

## 6 Conclusion

In this study, I compared a broad set of GARCH-type models with various GARCH specifications and innovation distribution assumptions in their empirical performance in one-step-ahead VaR and ES forecasting of four indices representing different asset classes. The goal has been to identify the best-performing specifications and distribution assumptions for each asset class, in order to identify the key characteristics they share and which are therefore essential for building a good financial risk forecasting GARCH-type model. Furthermore, I aimed to identify differences in the best-performing models between asset classes and shed light on the relative importance of the GARCH specification versus the distribution assumption. To achieve this, I used a three-step decision process, which firstly involves testing VaR and ES forecasts with two conditional coverage backtests to validate their accuracy. This is followed by ranking the models that have passed the backtesting based on a loss function for joint VaR and ES forecasting. Finally, I checked for significant differences in loss between the model with rank 1 and each subsequent model that has passed the backtesting. Through this process, I arrived at a set of best-performing models for each index. As a result, I can draw the following conclusions:

For both stock indices, GARCH specifications that include asymmetric terms lead to better results than their symmetric competitors. This advantage is more pronounced for

WIG than for DAX. Including kurtosis and skewness into the assumed distribution improves the model forecasts further, particularly for DAX. The GARCH specification tends to play a more important role for both stock indices. In this study, the distribution assumption rather ranked the modes within a GARCH specification. This appears to be the other way around for cryptocurrencies, represented by Bitcoin. Here, the distribution assumption seems to play a more important role than the GARCH specification. A distribution assumption which is capable of capturing leptokurtic innovations is crucial for a good model. A power parameter in the GARCH specification leads to small advantages over standard specifications. However, these advantages are often not significant. For Gold, the representative of commodities, the GARCH specification and distribution assumption appear to be equally important. The best-performing specifications either included a power parameter or a decomposition of volatility into a permanent and transitory component. The inclusion of kurtosis and, especially, skewness improve the model forecasts.

Overall, including an asymmetric term into the GARCH specification appears to be more important for stock indices than for cryptocurrencies and commodities. This is in line with the result from Hansen and Lunde (2005), who also found that the capability to model asymmetric volatility is particularly important for stock indices. A CCF plot can give a good hint on the importance of including asymmetric volatility into the GARCH specification. DAX and WIG both showed significant negative cross correlation coefficients in their log-return series between squared log-returns and lagged log-returns, while Bitcoin and Gold exhibit this feature either only slightly or not at all. For cryptocurrencies and commodities, the inclusion of a power parameter into the GARCH specification appears to be beneficial for good VaR and ES forecasts. Also models that decompose volatility into a permanent and transitory component seem to lead to superior forecasts for these assets. In terms of the distribution assumption, including excess kurtosis is important across all asset classes. Additionally also including skewness only improves forecasts for the stock indices and Gold.

A general statement about the relative importance of the GARCH specification versus the distribution assumption for a good model is hard to make. For the stock market indices, the GARCH specification is more influential on the performance of a model, likely because the ability to model asymmetric volatility is important there. For Bitcoin, the distribution assumption was more pivotal, while for Gold the relative importance of both components appears to be relatively equal.

The results of this study show that while certain characteristics of models appear to be more important for certain asset classes, definite conclusions about the overall best-performing

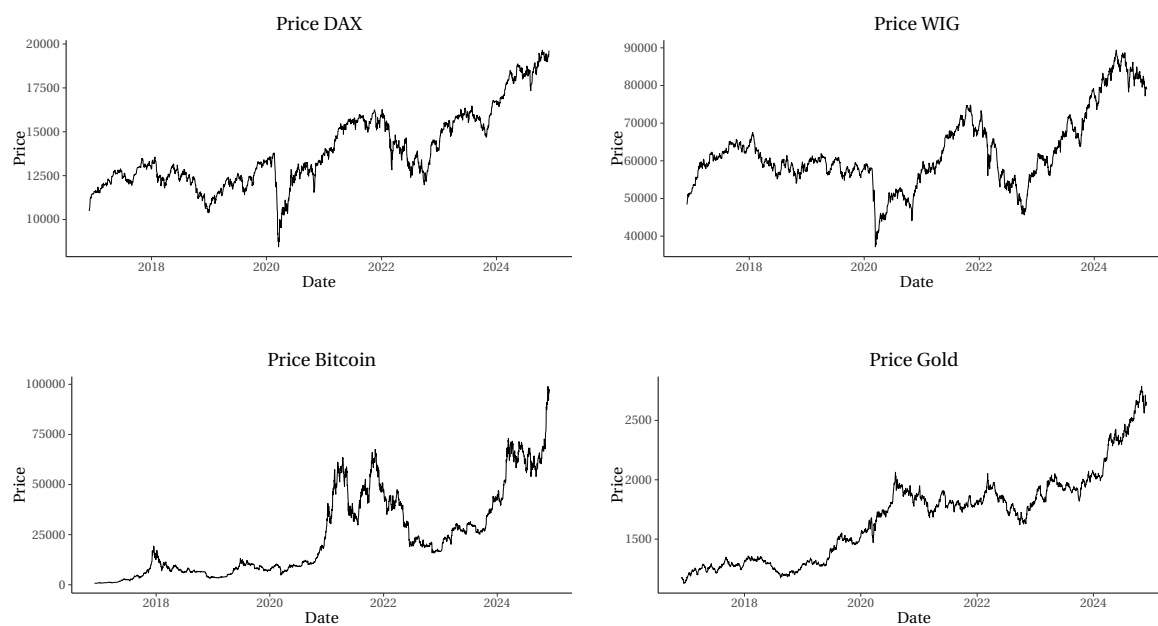
model cannot be made. For practical use, it is advisable to consider multiple models and determine the best-performing one for a given asset. The three-step decision process used in this study, could be employed for this purpose. The findings about important characteristics of GARCH specifications and distribution can help guide the selection of models to be considered for a particular asset. For further studies, it would be interesting to also consider more conservative tolerance levels and longer forecasting horizons.

Finally, Table 6.1 presents the top 10 best-performing models in this study for each index, along with the corresponding mean FZ loss and the  $p$ -value of the DM test, which compares each model to the model with rank 1. Plots of the models with rank 1 can be found in Figure A.6.

|    | DAX              |         |       | WIG              |         |       | Bitcoin       |         |       | Gold             |         |       |
|----|------------------|---------|-------|------------------|---------|-------|---------------|---------|-------|------------------|---------|-------|
|    | FZ Loss          | p value |       | FZ Loss          | p value |       | FZ Loss       | p value |       | FZ Loss          | p value |       |
| 1  | AVGARCH SGED     | -3.632  | 1.000 | FGARCH GHYP      | -3.585  | 1.000 | EGARCH GED    | -2.569  | 1.000 | CGARCH SGED      | -3.816  | 1.000 |
| 2  | NAGARCH SGED     | -3.631  | 0.916 | APARCH SSTD      | -3.585  | 0.961 | CGARCH SGED   | -2.567  | 0.859 | CGARCH GHYP      | -3.815  | 0.643 |
| 3  | GJR-GARCH SGED   | -3.630  | 0.910 | APARCH JSU       | -3.581  | 0.760 | CGARCH GED    | -2.567  | 0.817 | FGARCH SSTD      | -3.813  | 0.858 |
| 4  | NAGARCH JSU      | -3.629  | 0.803 | APARCH GHYP      | -3.581  | 0.706 | NGARCH STD    | -2.566  | 0.734 | CGARCH EMPIRICAL | -3.812  | 0.415 |
| 5  | NAGARCH SSTD     | -3.629  | 0.800 | APARCH STD       | -3.580  | 0.714 | APARCH GED    | -2.566  | 0.306 | NGARCH SGED      | -3.812  | 0.566 |
| 6  | NAGARCH GHYP     | -3.629  | 0.770 | FGARCH SSTD      | -3.578  | 0.303 | TGARCH STD    | -2.565  | 0.652 | FGARCH GHYP      | -3.812  | 0.763 |
| 7  | GJR-GARCH SSTD   | -3.628  | 0.825 | APARCH GED       | -3.577  | 0.553 | CGARCH GHYP   | -2.564  | 0.681 | NGARCH GHYP      | -3.810  | 0.445 |
| 8  | GJR-GARCH JSU    | -3.628  | 0.817 | APARCH EMPIRICAL | -3.577  | 0.517 | NGARCH GED    | -2.563  | 0.143 | CGARCH JSU       | -3.809  | 0.048 |
| 9  | FGARCH EMPIRICAL | -3.627  | 0.759 | APARCH SNORM     | -3.569  | 0.250 | EGARCH SGED   | -2.563  | 0.198 | APARCH GHYP      | -3.808  | 0.503 |
| 10 | GJR-GARCH GHYP   | -3.627  | 0.743 | APARCH NORM      | -3.568  | 0.322 | GJR-GARCH GED | -2.563  | 0.295 | NGARCH JSU       | -3.807  | 0.353 |

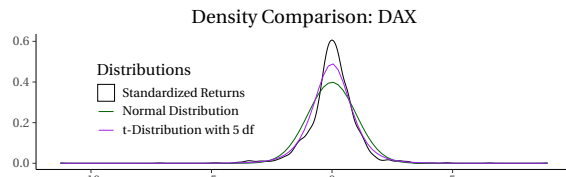
**Table 6.1: Top 10 Best-Performing Models based on FZ Loss**

## A Appendix

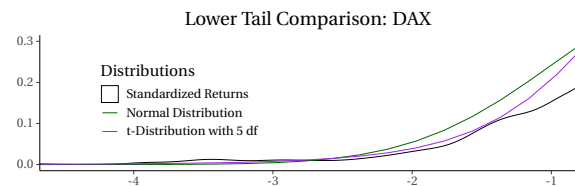


**Figure A.1: Price Plots**

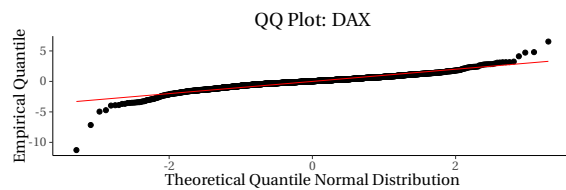
## Descriptive Plots of DAX Log-Returns



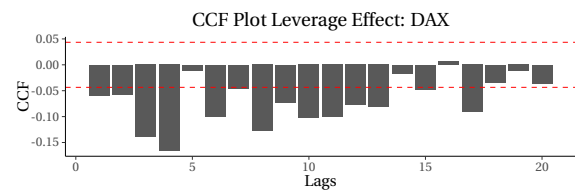
(a) Density of standardized log-returns compared to density of standard normal distribution and t-distribution with 5 degrees of freedom: Density of standardized log-returns has heavier tails compared to both other distributions and more weight in the center.



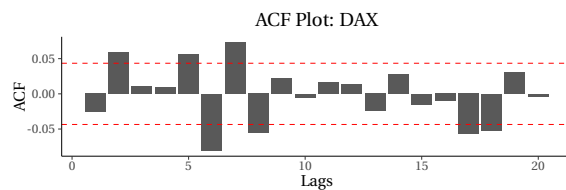
(b) Lower tail of standardized log-return density compared to lower tail of standard normal distribution and t-distribution with 5 degrees of freedom: More weight in outer tails of log-return density compared to both other distributions and less towards the center.



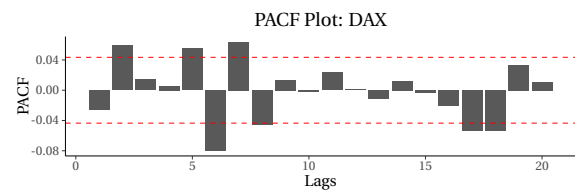
(c) QQ-Plot compares empirical quantiles with theoretical quantiles of the normal distribution: Discrepancies visible, especially in outer quantiles.



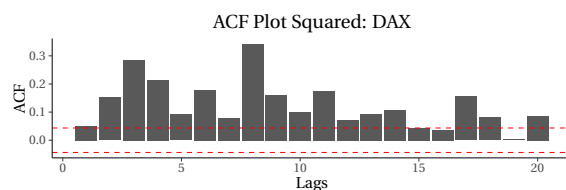
(d) Cross Correlation plot represents correlation between squared log-returns and lagged log-returns: Significant negative correlation coefficients indicate existence of leverage effect.



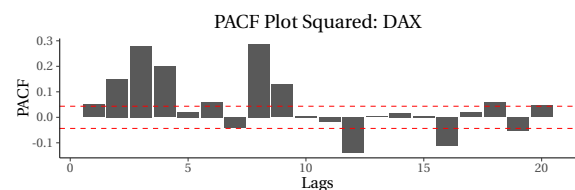
(e) Autocorrelation plot of log-returns: Slightly significant autocorrelation coefficients indicate minor, potentially neglectable serial autocorrelation in log-returns. These small autocorrelation coefficients could be a result of randomness.



(f) Partial Autocorrelation plot of log-returns: Slightly significant partial autocorrelation coefficients, similar to ACF, indicate minor, potentially neglectable, autocorrelation in log-return series. These small autocorrelation coefficients could be a result of randomness.



(g) Autocorrelation plot of squared log-returns: Significant autocorrelation coefficients across many lags indicate dependence of volatility on previous volatility and hence volatility clustering.

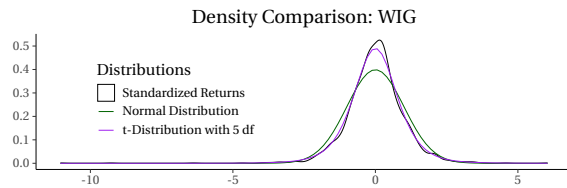


(h) Partial Autocorrelation plot of squared log-returns: Significant partial autocorrelation coefficients enhance suspicion of volatility clustering. Additionally, it can be seen that first nine lags are especially influential.

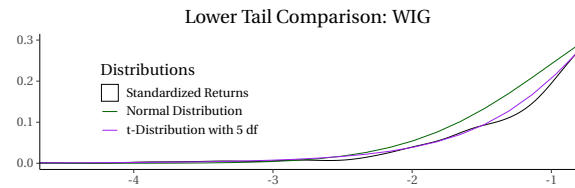
**Figure A.2: Descriptive Plots of DAX Log-Returns**



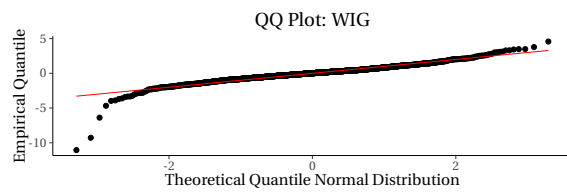
## Descriptive Plots of WIG Log-Returns



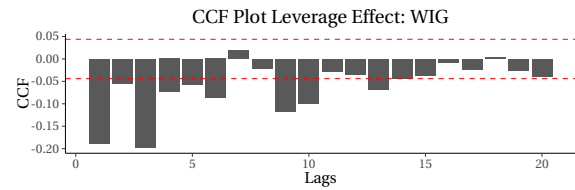
(a) Density of standardized log-returns compared to density of standard normal distribution and t-distribution with 5 degrees of freedom: Density of standardized log-returns has heavier tails compared to standard normal distribution and more weight in the center. Overall, it is well approximated by the t-distribution.



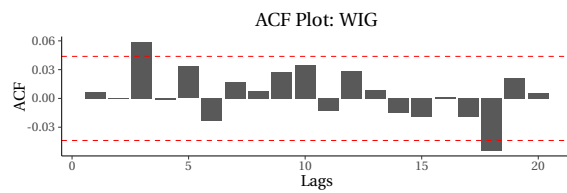
(b) Lower tail of standardized log-return density compared to lower tail of standard normal distribution and t-distribution with 5 degrees of freedom: More weight in outer tails of log-return density compared to standard normal distributions and less in the center. It is well approximated by the t-distribution.



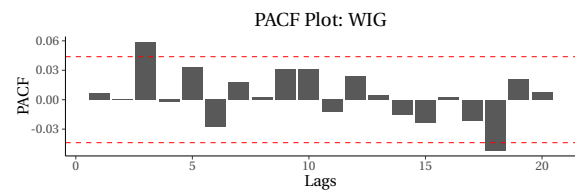
(c) QQ-Plot compares empirical quantiles with theoretical quantiles of the normal distribution: Discrepancies visible, especially in outer quantiles.



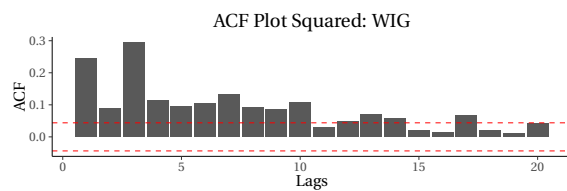
(d) Cross Correlation plot represents correlation between squared log-returns and lagged log-returns: Significant negative correlation coefficients indicate existence of leverage effect.



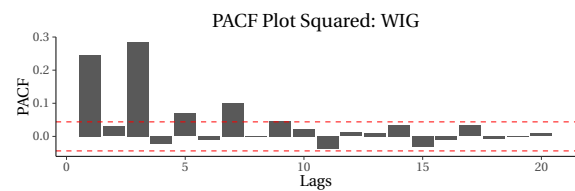
(e) Autocorrelation plot of log-returns: Only one significant autocorrelation coefficient in third lag indicates that log-returns are probably not autocorrelated.



(f) Partial Autocorrelation plot of log-returns: Only one significant partial autocorrelation coefficient in third lag indicates that log-returns are probably not autocorrelated.



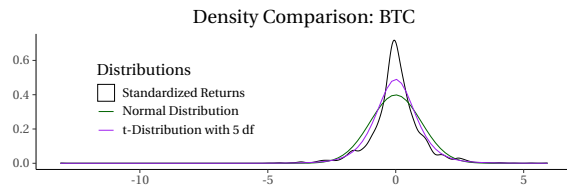
(g) Autocorrelation plot of squared log-returns: Significant autocorrelation coefficients across many lags indicate dependence of volatility on previous volatility and hence volatility clustering.



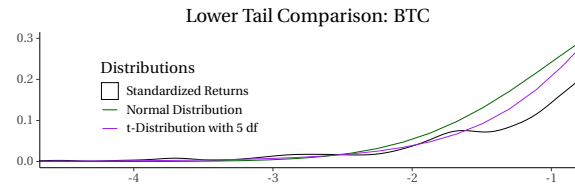
(h) Partial Autocorrelation plot of squared log-returns: Significant partial autocorrelation coefficients enhance suspicion of volatility clustering. Additionally, it can be seen that first seven lags are especially influential.

**Figure A.3: Descriptive Plots of WIG Log-Returns**

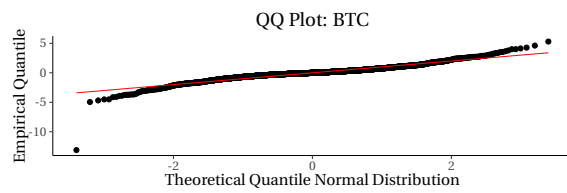
## Descriptive Plots of Bitcoin Log-Returns



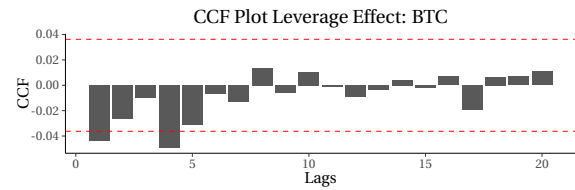
(a) Density of standardized log-returns compared to density of standard normal distribution and t-distribution with 5 degrees of freedom: Density of standardized log-returns has heavier tails compared to both other distributions and more weight in the center.



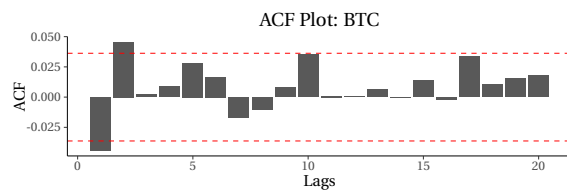
(b) Lower tail of standardized log-return density compared to lower tail of standard normal distribution and t-distribution with 5 degrees of freedom: More weight in outer tails of log-return density compared to both other distributions and less towards the center.



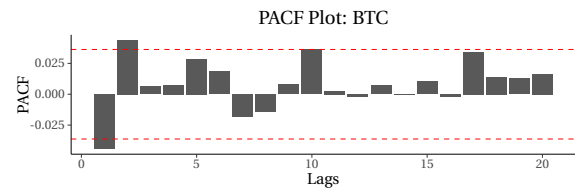
(c) QQ-Plot which compares empirical quantiles with theoretical quantiles of normal distribution: Discrepancies visible, especially in outer quantiles.



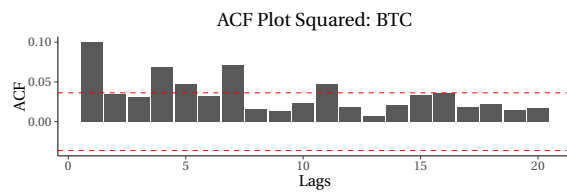
(d) Cross Correlation plot represents correlation between squared log-returns and lagged log-returns: Weak and slightly significant autocorrelation coefficients in third and fourth lags. Existence of weak leverage effect possible but not very obvious.



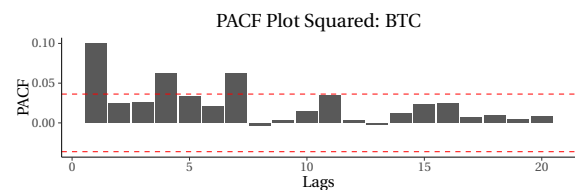
(e) Autocorrelation plot of log-returns: First two autocorrelation coefficients are weakly significant. This can indicate slight serial autocorrelation in log-returns. However, there is no clear sign for it and the significant autocorrelation coefficients could be due to randomness.



(f) Partial Autocorrelation plot of log-returns: As in ACF, first two autocorrelation coefficients are weakly significant. This can indicate slight serial autocorrelation in log-returns, but it is no clear sign for it.



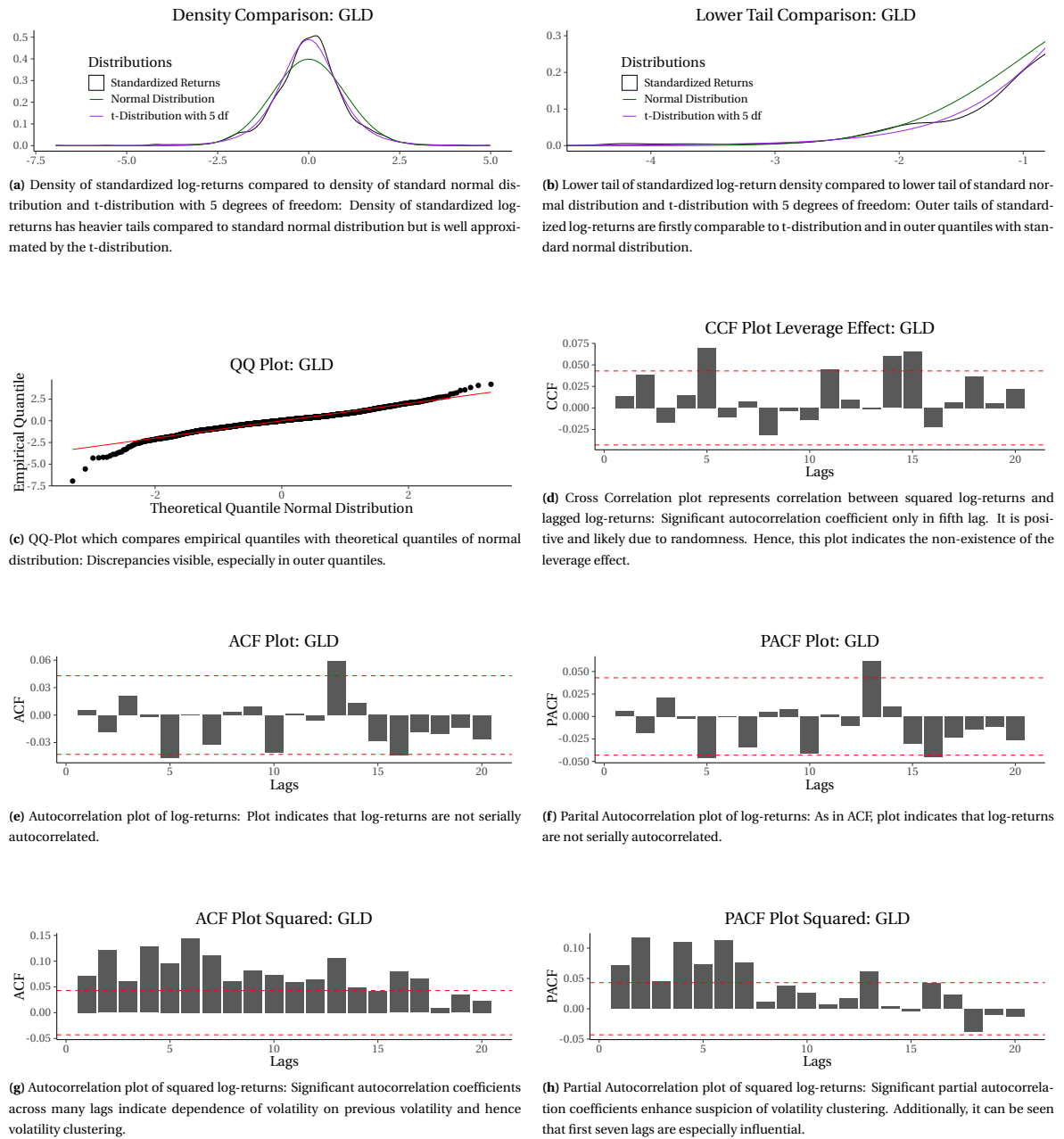
(g) Autocorrelation plot of squared log-returns: Some significant autocorrelation coefficients across first seven lags indicate dependence of volatility on previous volatility and hence volatility clustering. However, this dependence is weaker than in stock market indices.



(h) Partial Autocorrelation plot of squared log-returns: Some significant partial autocorrelation coefficients in first seven lags enhance suspicion of volatility clustering, although a weaker one than in stock market indices.

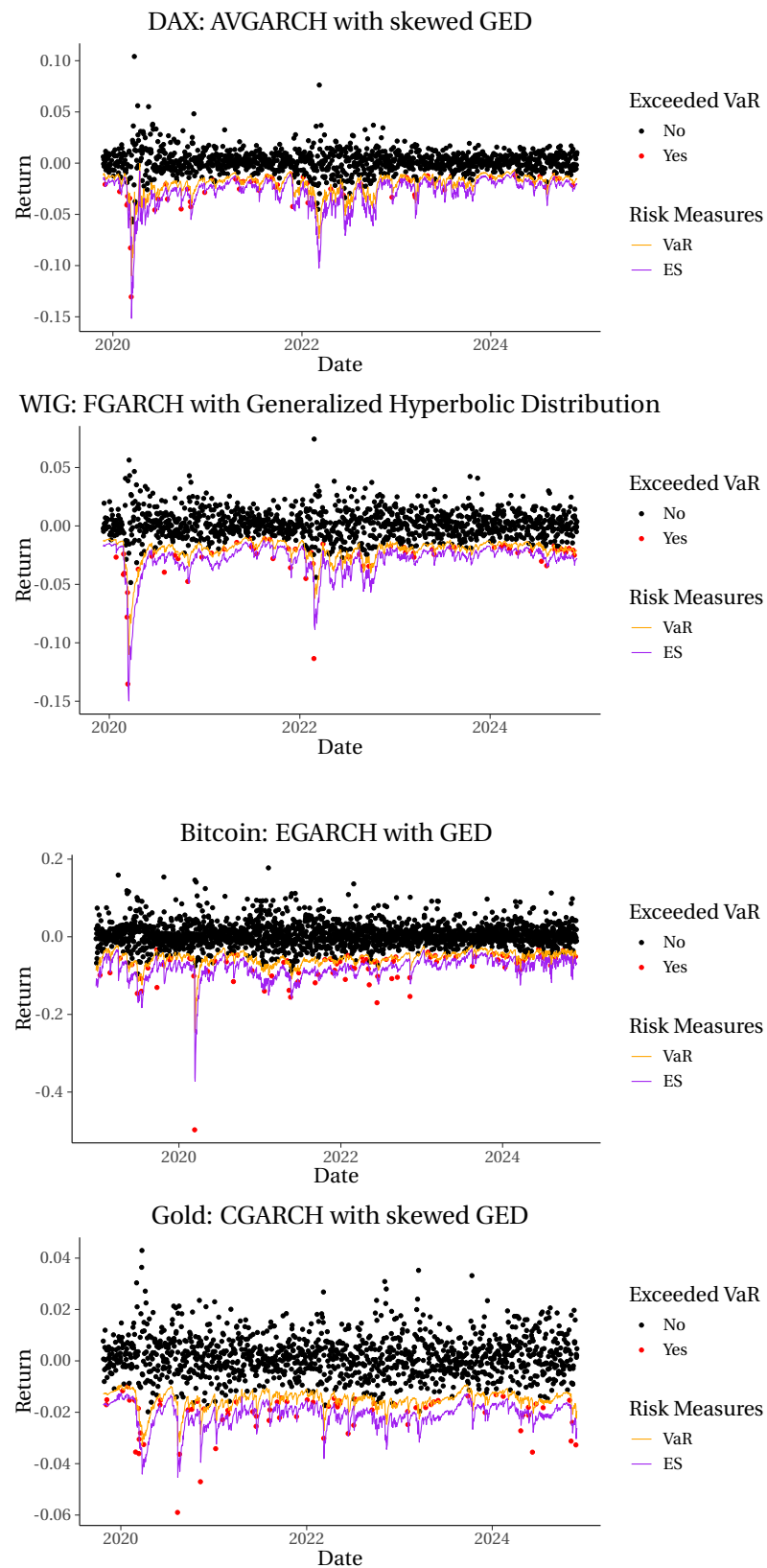
**Figure A.4: Descriptive Plots of Bitcoin Log-Returns**

## Descriptive Plots of Gold Log-Returns



**Figure A.5: Descriptive Plots of Gold Log-Returns**

## VaR and ES Forecasts vs Log-Returns: Models with Rank 1



**Figure A.6: VaR and ES Forecasts vs Log-Returns: Models with Rank 1**

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## **Erklärung**

Ich versichere, dass ich die Arbeit selbständig und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt habe. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen oder anderen Quellen entnommen sind, sind als solche kenntlich gemacht. Die schriftliche und elektronische Form der Arbeit stimmen überein. Ich stimme der Überprüfung der Arbeit durch eine Plagiatssoftware zu.

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